BENCHMARKING MOTION DEBLURRING UNDER OPTICAL IMAGE STABILIZATION USING A TRANSFORMER-BASED MODEL

An Undergraduate Thesis Outline
Submitted to the Faculty of
College of Engineering and Information Technology
Cavite State University
Indang, Cavite

In partial fulfilment
of the requirements for the degree of
Bachelor of Science in Computer Science

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February 2026

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF APPENDICES

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An undergraduate thesis outline submitted to the faculty of the Department of Information Technology, College of Engineering and Information Technology, Cavite State University, Indang, Cavite in partial fulfillment for the degree of Bachelor of Science in Computer Science with Contribution No. ________________. Prepared under the supervision of Prof. Ria Clarisse M. Sy

INTRODUCTION

Photography plays a vital role in modern life by allowing individuals to document experiences, preserve meaningful moments, and obtain visual information. Despite advances in imaging technology, achieving clear image outputs remains a challenge. One of the most common issues in image acquisition is motion blur. Motion blur occurs when the camera or the subject moves during image capture, resulting in degraded image quality (Zeng & Diao, 2020).

Motion blur significantly reduces the visual clarity and practical usefulness of images. While blur may be intentionally applied for artistic purposes, it is generally undesirable in technical and real-world applications. In computer vision tasks such as optical character recognition (OCR), face recognition, and object detection, accurate visual details are critical for reliable performance (Tomionko, Traoré, & Traoré, 2022; Shi & Tan, 2023; K C, M, & Rafi, 2022).

Image restoration is an important area of computer science that focuses on recovering and enhancing degraded images, including those affected by motion blur (Kumar, Mastan, & Raman, 2022). Among various image restoration tasks, motion deblurring is widely recognized as one of the most challenging because motion blur degrades visual quality for human perception and simultaneously disrupts the extraction of reliable features required for computer vision tasks due to its complex and scene-dependent characteristics (Arslan, Gultekin, & Saranli, 2024). Recent advances in machine learning, particularly deep learning–based approaches, have shown improved performance in motion deblurring by learning restoration patterns directly from large datasets (Xiang et al., 2024).

In mobile photography, hardware-based stabilization techniques such as Optical Image Stabilization (OIS) are widely integrated into smartphones to reduce camera shake during handheld image capture. OIS compensates for pitch and yaw movements through motor-driven lens adjustments, helping to reduce motion-induced blur, especially in low-light or unstable conditions (Rosa, Virzi, Bonaccorso, & Branciforte, 2015). While OIS is designed to reduce blur at the image acquisition stage, its influence on the characteristics of motion blur captured by the camera is not always explicitly considered in the evaluation of learning-based deblurring models.

Current benchmarking practices in motion deblurring present additional challenges. Many existing benchmarks rely heavily on synthetic blur because collecting paired real-world blurred and sharp images is difficult and time-consuming. As noted by Rim, Lee, Won and Cho (2020), the absence of datasets consisting of real-world blurred images and corresponding ground-truth sharp images has forced most learning-based approaches to depend on synthetic data, which often results in poor generalization to real-world motion blur. In addition, some real-world datasets exhibit limited diversity in capture devices. One such case is the dataset presented by Mahmud et al. (2025), in which all videos were recorded using a single smartphone model, which may limit generalization to other hardware.The lack of datasets that systematically capture image pairs acquired both with OIS and without OIS, each consisting of corresponding sharp and motion-blurred images from the same scene, reveals a lack of comprehensive benchmarking resources in current motion deblurring research. This limitation restricts the ability to evaluate motion deblurring models under realistic hardware stabilization conditions and to examine how such conditions influence restoration performance. Addressing this limitation is essential for developing more reliable and generalizable motion deblurring benchmarks that better reflect real-world image acquisition settings.

In response to this gap, the present study intends to construct a real-world, stabilization-aware benchmark and systematically evaluate motion deblurring performance under both OIS and non-OIS capture conditions using a transformer-based model.

Statement of the Problem

Many existing motion deblurring benchmarks rely on synthetic blur because collecting paired degraded and undegraded images from real-world scenes is often infeasible. As a result, available real-world datasets tend to have limited scene coverage and insufficient diversity in real degradation patterns. This limitation provides weak priors for learning-based motion deblurring models and leads to notable performance degradation when trained models are applied to real-world images (Zhai, Wang, Cui, & Zhou, 2023). Moreover, when real-world datasets are available, they are commonly captured using high-end imaging devices, such as mirrorless cameras, which do not accurately reflect the imaging conditions of devices most widely used in practice, particularly smartphones (Rim et al., 2020). This reliance on high-end cameras limits the applicability of benchmarking results to motion deblurring scenarios encountered in everyday mobile photography.

In the context of mobile imaging, OIS has become a standard feature in modern smartphones, as it reduces motion blur caused by natural hand tremors during handheld image capture through hardware-level lens adjustments (Liu et al., 2024). While OIS improves image quality in everyday photography, it also presents a challenge for motion deblurring research by reducing the occurrence of naturally motion-blurred images in real-world datasets. Consequently, there is insufficient data and limited comparative analysis examining how motion deblurring models perform on images captured with OIS and those captured without OIS (Li, 2024; Mahmud et al., 2025).

Current benchmarking practices also commonly evaluate motion deblurring models using publicly available pre-trained models without further adaptation to the target dataset. As noted by Mahmud et al. (2025), although this practice highlights generalization gaps, it may not reflect the full performance potential of each model when applied to specific real-world data. The absence of dataset-specific fine-tuning limits the ability of existing benchmarks to assess model performance under realistic capture conditions and after domain adaptation.

In addition, existing motion deblurring studies are largely dominated by foreign research, with limited locally published studies focusing on image deblurring. In particular, no local study was identified that systematically benchmarks motion deblurring models using images captured with and without OIS. This situation highlights the lack of local empirical evidence that reflects practical imaging conditions relevant to commonly used mobile devices.

At present, to the best of the researchers' knowledge, no publicly available datasets specifically focus on smartphone-acquired images that systematically provide corresponding image pairs captured both with OIS and without OIS. In particular, there is no dataset that ensures each pair contains sharp and motion-blurred images of the same scene under controlled mobile acquisition settings (Shibu, 2023; Xiang, 2024). The absence of such datasets restricts the ability to conduct systematic and fair benchmarking of motion deblurring models under varying hardware stabilization conditions.

Given these limitations in existing datasets and benchmarking practices, there remains a lack of systematic evaluation of motion deblurring performance under real-world with OIS and without OIS conditions. This gap limits understanding of how hardware-based stabilization influences image restoration performance and the generalization behavior of a deblurring model.

Objectives of the Study

This study aims to systematically benchmark motion deblurring performance under different hardware stabilization conditions (with OIS and without OIS) using a transformer-based model, and to evaluate the influence of domain-adaptive fine-tuning on real-world deblurring performance. Specifically, the study seeks to:

1. Construct a curated dataset of paired real-world images consisting of motion-blurred and sharp images captured under different hardware stabilization conditions (with OIS and without OIS), to serve as a realistic benchmark for stabilization-aware motion deblurring.

2. Evaluate the baseline restoration performance of a pre-trained transformer-based model on the curated dataset across with OIS and non-OIS capture conditions using quantitative image quality metrics, namely Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM).

3. Compare the restoration performance of the transformer-based model after domain-adaptive fine-tuning on the constructed dataset with the pre-trained

baseline to examine whether OIS-aware real-world data improves motion deblurring performance.

4. Analyze the statistical significance of performance differences between (a) images captured with OIS and without OIS, and (b) outputs produced by the pre-trained and fine-tuned models, to quantify the impact of hardware stabilization and domain-adaptive fine-tuning on motion deblurring performance.

Hypothesis

This study will test hypotheses related to the effects of hardware-based stabilization and domain-adaptive fine-tuning on motion deblurring performance in mobile photography. Restoration performance will be evaluated using PSNR and SSIM.

Hypothesis 1: Effect of OIS on Restoration Performance

Null Hypothesis (H_{01}). There is no statistically significant difference in motion deblurring performance, as measured by PSNR and SSIM, between images captured with OIS and images captured without OIS.

Alternative Hypothesis (H_{11}). There is a statistically significant difference in motion deblurring performance, as measured by PSNR and SSIM, between images captured with OIS and images captured without OIS.

Hypothesis 2: Effect of Fine-Tuning on Model Performance

Null Hypothesis (H_{02}). Domain-adaptive fine-tuning of the pre-trained transformer-based model does not result in a statistically significant improvement in motion deblurring performance compared to the pre-trained baseline.

Alternative Hypothesis (H_{12}). Domain-adaptive fine-tuning of the pre-trained transformer-based model results in a statistically significant improvement in motion deblurring performance compared to the pre-trained baseline.

Significance of the Study

This study is significant as it addresses key gaps in image restoration research and mobile photography by providing a real-world benchmark dataset containing paired motion-blurred and sharp images captured with OIS and without OIS. Such datasets are largely absent in existing studies, limiting the ability to systematically evaluate how hardware stabilization influences motion blur characteristics and restoration performance. By enabling controlled comparisons across stabilization settings, the proposed dataset is expected to support more realistic and generalizable benchmarking of motion deblurring models under practical mobile imaging settings.

In addition, the study contributes methodologically by benchmarking both pre-trained and domain-adaptively fine-tuned transformer-based deblurring models. Existing benchmarking practices commonly evaluate pre-trained models without dataset-specific adaptation, which may restrict understanding of their performance on real-world data. By incorporating fine-tuning and statistical significance testing, this study is expected to provide empirical evidence on the impact of real-world data adaptation and hardware stabilization conditions on restoration outcomes.

The findings of this study are expected to be relevant to researchers, system developers, and industry practitioners involved in computational photography and computer vision. Improved understanding of the effects of OIS on motion deblurring performance may inform the design, evaluation, and deployment of software-based image restoration algorithms and imaging pipelines for mobile and embedded systems. These contributions align with Sustainable Development Goal 9: Industry, Innovation, and Infrastructure, by supporting the advancement of evaluation practices and technologies that strengthen digital imaging systems and innovation-driven industries.

Finally, the study aligns with the Cavite State University (CvSU) Thematic Area of Smart Engineering, Information and Communication Technology (ICT), and Industrial Competitiveness. Through the development of a curated real-world dataset and systematic benchmarking procedures, the study aims to contribute empirical evidence that supports ICT-related research and engineering solutions. These outcomes are expected to promote the development of more reliable and competitive smart imaging applications relevant to commonly used mobile devices.

Scope and Limitations of the Study

This study will focus on evaluating the influence of OIS on the performance of a transformer-based motion deblurring model, specifically the Multi-scale Network with Learnable Discrete Wavelet Transform (MLWNet), within the context of mobile photography. The investigation will be limited to motion blur generated through controlled motion strategies applied within real-world image capture conditions. These strategies will include hand-induced shake, vibration-based motion, and controlled rig-assisted motion designed to produce approximately linear motion blur.

Other forms of image degradation such as defocus blur will not be treated as primary variables and will not be separately evaluated. Sensor noise may be present due to real-world image capture; however, its effect is minimal under well-lit conditions. Compression artifacts may also be introduced during image conversion or storage. These factors may slightly influence pixel-based evaluation metrics.

Consistent preprocessing and evaluation procedures will be applied across all samples to maintain fair comparison.

Image capture will be conducted under normal to well-lit conditions. Extreme low-light conditions requiring long exposure times or very high ISO settings will be avoided, although moderate lighting variations may still occur and are not treated as a controlled variable.

The dataset will consist of real-world images of predominantly static scenes where motion blur is primarily induced by controlled camera movement, captured using mobile devices with and without OIS. Minor environmental motion, such as slight movement of leaves or grass, may be present but is considered negligible.

Scenes containing readable text, printed characters, or textual patterns will not be included to avoid task-specific bias and to ensure that evaluation focuses solely on general visual restoration quality rather than text legibility or optical character recognition performance. For each scene, corresponding sharp and motion-blurred image pairs will be collected using the defined motion strategies to enable controlled comparison across stabilization settings.

Two smartphone models from the same manufacturer will be used consistently throughout data collection, with one device equipped with OIS and the other without OIS. Selecting devices from the same manufacturer is intended to reduce variability caused by brand-level hardware and software differences, thereby allowing clearer isolation of the stabilization mechanism as the primary variable of interest. Although the devices may differ in certain hardware characteristics such as sensor size and pixel structure, standardized capture settings will be applied across both devices, including identical resolution, frame rate, manual exposure parameters, and disabled electronic image stabilization features. These controls will be implemented to minimize confounding factors unrelated to OIS. However, complete isolation of OIS as a single variable cannot be guaranteed due to inherent hardware differences between devices.

Variations in color reproduction between the two mobile devices may occur due to differences in camera sensors and imaging pipelines, even when using RAW images. To reduce this effect, a basic color alignment procedure will be applied to improve consistency between image pairs. However, full color calibration is beyond the scope of the study, as the focus is on motion deblurring rather than color fidelity. A consistent preprocessing pipeline and structural evaluation metrics (PSNR and SSIM) will be used, although minor color-related discrepancies may still introduce limited bias.

A structured paired image post-processing pipeline will be applied to ensure spatial and photometric consistency between sharp and motion-blurred image pairs. Geometric alignment will be performed to achieve pixel-level correspondence between paired frames through spatial transformation. It is acknowledged that this alignment process may introduce minor distortions and slight misregistration due to imperfect transformation. Following alignment, resampling will be applied using Bicubic interpolation to standardize image grids. This interpolation step may introduce smoothing and intensity variations, particularly in high-frequency regions. These artifacts cannot be fully avoided and may slightly affect pixel-level features that are sensitive to transformer-based models. To address this, a consistent alignment and interpolation strategy will be applied across all samples to reduce bias and ensure fair comparison between conditions.

The study is designed to examine the relative influence of OIS under controlled mobile imaging conditions rather than to establish broad cross-device generalization. Consequently, the findings may not fully represent motion blur behavior across other smartphone brands, camera architectures, or professional imaging systems such as digital single-lens reflex (DSLR) platforms. The exclusive focus on inanimate and non-textual scenes further limits applicability to scenarios involving human subjects, facial content, text-centric imagery, or complex dynamic motion.

Since images will be captured in RAW format, the influence of proprietary image processing algorithms will be minimized. However, physical differences between the camera systems may still influence the severity and pattern of motion blur. Variations in motion intensity during capture, particularly in the hand-induced shake and rig-assisted motion setups, may lead to differences in blur characteristics because the magnitude and consistency of movement cannot be controlled. Vibration-based motion is expected to produce relatively more consistent blur due to its more controlled mechanical behavior.

The evaluation will be limited to a single transformer-based deblurring architecture. The study will assess both the pre-trained and domain-adaptively fine-tuned versions of MLWNet using quantitative image quality metrics, specifically PSNR and SSIM. A two-way ANOVA will be employed to determine the significance of performance differences between stabilization conditions and training configurations. Observed trends may differ when alternative architectures or traditional deblurring methods are used.

No human subjects, facial data, or personally identifiable information (PII) will be included in the dataset.

Definition of Terms

Benchmarking refers to the systematic evaluation and comparison of artificial intelligence models using standardized datasets, metrics, and experimental protocols to assess performance and limitations (Reuel et al., 2024).

Curated dataset refers to a structured and documented collection of images prepared and maintained to support reliable model training and evaluation (Gomstyn & Jonker, 2025).

Fine-tuning refers to the process of adapting a pre-trained machine learning model to a specific task or dataset through additional training (Bergmann, 2025).

Image restoration refers to the process of improving image quality by recovering a sharp image from a degraded input such as blur or noise (Reeves, 2014).

Motion blur refers to image degradation caused by camera or subject movement during exposure, resulting in streaked or smeared regions (Zeng & Diao, 2020).

Motion deblurring refers to a computer vision technique that restores sharp image details from motion-blurred images (Huihui, Daoliang, & Yingyi, 2023).

Optical Image Stabilization (OIS) refers to a hardware-based method that reduces blur by compensating for camera motion through lens adjustments before image capture (Liu et al., 2024).

Peak Signal-to-Noise Ratio (PSNR) refers to a pixel-based image quality metric derived from mean squared error that measures numerical similarity between restored and reference images (Hore & Ziou, 2010).

Pre-trained model refers to a machine learning model trained on a large dataset and reused or fine-tuned for related tasks (Stryker, 2025).

Quantitative metrics refers to numerical measures used to objectively evaluate and compare model performance.

Real-world blur refers to motion blur produced during actual image capture, influenced by camera motion, sensor noise, and image signal processing (Rim et al., 2022).

Restoration performance refers to the effectiveness of an image restoration or motion deblurring model as measured using quantitative metrics.

Sharp reference refers to the ground-truth, blur-free image used to evaluate the accuracy of restored images (Nah, Kim, & Lee, 2017).

Statistical significance refers to a statistical result indicating that observed differences are unlikely to have occurred by chance (Tenny & Abdelgawad, 2023).

Structural Similarity Index Measure (SSIM) refers to a perceptual image quality metric that evaluates similarity based on luminance, contrast, and structural information (Wang, Bovik, Sheikh, & Simoncelli, 2004).

Synthetic blur refers to artificially generated blur created from sharp images under controlled conditions and typically less complex than real-world blur (Rim et al., 2022).

Transformer refers to a neural network architecture originally designed for sequential data that has demonstrated strong performance in computer vision and other artificial intelligence tasks (Stryker & Bergmann, 2025).

RELATED LITERATURE

This chapter presents a structured review of theories, literature, and empirical studies related to motion deblurring in digital imaging. It establishes the conceptual and research foundations necessary to understand motion blur, image restoration, and deep learning–based deblurring methods. The chapter also examines existing benchmarking practices, real-world datasets, and hardware stabilization mechanisms, with particular emphasis on OIS. Through this review, limitations in current research are identified to justify the need for systematic benchmarking under OIS and non-OIS conditions.

Theoretical Background

The theoretical background presents the fundamental concepts and principles related to image restoration and motion deblurring. It covers image degradation mechanisms, the formation and modeling of motion blur, classical and learning-based deblurring approaches, deep learning architectures, and image quality assessment metrics. These concepts establish the technical foundation for understanding how motion blur occurs, how it can be addressed, and how restoration performance is evaluated.

Image Restoration in Computer Vision

Definition and Purpose of Image Restoration. Image restoration is a critical branch of image processing focused on recovering degraded images to enhance visual quality and utility. While image processing encompasses a broad range of techniques for analyzing, enhancing, and manipulating images, image restoration specifically aims to reconstruct high-quality images from degraded captures (Wali, Naseer, Tamoor, & Gilani, 2023; Zuo, 2025).

Restoration serves two primary purposes: improving pictorial information for human interpretation and preparing image data for storage, transmission, or automated machine perception. It restores important image details that are often lost during capture or post-processing, including those affected by compression or transmission (M, Mg, & My, 2024).

Impact of Image Degradation Across Application Domains. The impact of image degradation extends across multiple domains. In computer science, degraded inputs reduce the performance of machine learning models; for instance, motion blur, haze, and underwater distortions have been shown to significantly lower Convolutional Neural Network (CNN) classification accuracy (Pei, Huang, Zou, Zhang, & Wang, 2019).

In medical imaging, noise and acquisition artifacts hinder the detection and diagnosis of diseases in modalities such as Magnetic Resonance Imaging (MRI), Optical Coherence Tomography (OCT), and ultrasound, which are particularly sensitive to low contrast and small anatomical structures. Effective image restoration is therefore essential for accurate downstream analysis (Sahu, Singh, Agrawal, & Wang, 2023).

In remote sensing, hyperspectral image (HSI) classification is a fundamental task widely used in agricultural production, forestry protection, mineral exploration, urban and rural planning, and land management. However, real-world hyperspectral images often suffer from degradation such as low spatial resolution, noise, fog, and shadow, which disrupt both spectral and spatial features and significantly reduce classification accuracy (Li, Li, Liu, &, Li, 2022).

Image degradation also affects intelligent transportation systems, where adverse weather conditions such as haze, rain, and snow degrade image and video quality. These degradations reduce detection reliability and increase errors in applications such as autonomous driving and traffic monitoring (Galshetwar et al., 2025).

Similarly, in Internet of Things (IoT)–based vision applications, images captured in outdoor environments are often affected by environmental conditions and camera-related factors such as lighting variations and lens distortions. These degradations negatively impact the performance of deep learning models, highlighting the need to consider both environmental and hardware influences when developing robust computer vision systems (Kaur et al., 2023).

These findings show that effective image restoration is important for maintaining reliable performance and accurate results in different computer vision applications.

Sources and Types of Image Degradation. Image degradation can result from capture conditions, environmental factors, or hardware limitations (Wali et al., 2023). Common distortions include noise, geometric artifacts (e.g., pincushion distortion), illumination and color inconsistencies, and blur (Lagendijk & Biemond, 2009). Recent studies further identify additional sources, including Gaussian and real-world noise, motion blur, defocus, low-light artifacts, JPEG compression, mosaic artifacts, and limitations imposed by camera hardware (Jiang, Hu, & Jia, 2024).

Image Restoration Tasks in Computer Vision. Single-degradation image restoration (SDIR) targets images affected by a specific type of degradation and typically involves task-specific models designed for that particular issue (Jiang et al., 2024). Low-level vision restoration techniques are commonly categorized according to degradation type, including super-resolution (SR), denoising, deraining, dehazing, deblurring, and color correction (Zhai et al., 2023).

Overview of Blur and Motion Blur

Blur as an Image Degradation Phenomenon. Blur is a common form of image degradation that reduces image sharpness and weakens the clarity of visual information. It may result from several factors, including camera shake, subject motion, inaccurate focus, insufficient depth of field, and lens limitations (Frye, 2020). Based on spatial extent, blur can be classified as global, where the entire image is affected due to camera movement, or local, where only moving objects appear blurred while the background remains relatively sharp (Hsu & Chen, 2008).

Types of Blur in Digital Imaging. Blur appears in different forms depending on its physical cause. Gaussian blur is commonly used as a smoothing operation that reduces noise and fine details by averaging pixel values using a Gaussian function (Akbar & Verma, 2024). Defocus blur occurs when light rays from a scene point fail to converge on the image sensor because the subject lies outside the focal plane. Its severity depends on lens aperture, object distance, focal depth, and sensor resolution (Joshi, 2020). While these blur types affect image clarity, motion blur remains one of the most challenging forms of degradation in computer vision.

Formation and Characteristics of Motion Blur. Motion blur occurs when the camera or subject moves during exposure, causing smeared or streaked regions in the captured image (Zeng & Diao, 2020). Welander, Marsh, & Gruber (2023) discussed that this effect arises when the shutter speed of the camera is too slow relative to the motion of the camera or objects in the scene, allowing the sensor to continuously collect light information while the object is moving. This idea implies that the degree of motion blur depends on the relationship between camera movement and shutter speed. Rapid camera movement captured using a slow shutter speed tends to produce strong motion blur, whereas minimal camera movement recorded with a fast shutter speed generally results in sharper images. When the shutter speed is insufficient to compensate for camera-induced motion, the sensor records changes in position over time, which produces the characteristic blurred appearance. According to Joshi (2020), motion blur represents the projection of an object's motion path onto the image plane, which may result from translational or rotational movement of either the camera or scene objects. In low-light conditions that require longer exposure times, these motion paths become more pronounced, leading to more severe blur.

Mathematical Modeling of Motion Blur. Motion blur can be described through the image formation process, in which each image pixel records light from multiple scene points during the exposure time. According to Wang and Tao (2014), this integration is influenced by both extrinsic factors, such as camera or object motion, and intrinsic factors, such as lens imperfections and defocus. For image restoration purposes, the full image formation pipeline is commonly simplified by focusing on the blur effect at the image plane.

Under this simplified assumption, blur is modeled as the convolution of a latent sharp image with an effective blur kernel, followed by the addition of noise:

$$y = x * h + n$$

where y is the observed blurred image, x is the latent sharp image, h denotes the blur kernel or point spread function (PSF), and n represents noise. The kernel h is treated as a combined representation of both extrinsic and intrinsic blur components (Wang & Tao, 2014).

This formulation is widely adopted due to its mathematical simplicity and serves as the foundation for most image deblurring methods. However, this formulation commonly assumes a spatially invariant blur kernel, an assumption that is mainly applicable to a limited range of camera motions. Joshi, Kang, and Szeliski (2014) explained that blur caused by camera shake often results from complex and general camera motion during long exposure, which cannot always be represented by a single, uniform blur kernel. In such cases, the blur kernel may vary spatially across the image, making the blind deconvolution problem more ill-posed and limiting the effectiveness of traditional deblurring methods that rely on spatial invariance.

Global and Local Motion Blur. Motion blur in real-world images can be further categorized into global and local types based on the spatial behavior of the blur. Global motion blur is primarily caused by camera shake and affects the entire image, resulting in relatively uniform degradation across the scene. In contrast, local motion blur arises from object motion, depth variation, or out-of-focus regions within a scene, producing blur that is confined to specific areas and varies spatially. Studies on real dynamic scenes have shown that these two forms of blur often coexist and are randomly coupled, with background regions exhibiting global blur and foreground moving objects showing localized blur. This coupling leads to non-uniform blur patterns that cannot be accurately represented by a single blur kernel, further increasing the difficulty of blind image deblurring and limiting the effectiveness of conventional model-based approaches (Li, Yang, Huang, & Zhuo, 2021).

Effects of Motion Blur in Real-World Vision Systems. Motion blur significantly degrades the performance of real-world vision systems. In robotics and autonomous navigation, rapid camera motion and dynamic environments reduce the reliability of feature extraction and visual odometry, leading to localization errors and tracking failures (Luan, Yang, Qin, Chen, & Sui, 2024). In intelligent transportation systems, high vehicle speeds combined with low illumination conditions introduce severe motion blur that degrades object detection performance in advanced driver assistance systems and autonomous driving applications (Zhang et al., 2023).

Similar challenges are observed in video surveillance, visual tracking, and precision agriculture, where camera motion and moving targets produce complex blur patterns that reduce detection accuracy and system reliability (Ma et al., 2016; Huihui et al., 2023).

These observations highlight that motion blur remains a persistent and challenging degradation in real-world image acquisition. Its dependence on camera motion, exposure settings, scene dynamics, and hardware characteristics underscores the importance of evaluating motion deblurring methods under realistic capture conditions.

Machine Learning Paradigm for Deblurring

Artificial Intelligence (AI) focuses on developing systems capable of performing tasks that normally require human intelligence, while machine learning enables computers to learn patterns directly from data. In computer vision, these technologies are widely used to analyze, interpret, and enhance images. Together, they form the foundation of modern image restoration techniques (Wu et al., 2025).

Recent advancements in AI and machine learning have enabled computational systems not only to approximate human visual perception but also to perform complex image analysis tasks that exceed the capabilities of traditional algorithms. Unlike classical methods that depend on manually designed rules, deep learning models automatically learn hierarchical visual features through large-scale, data-driven training. This approach has led to significant improvements in accuracy and adaptability. As a result, deep neural networks have demonstrated superior performance in tasks such as object detection, semantic segmentation, and various image enhancement problems (Chen, Wang & Hsia, 2025).

These advances have driven progress across multiple application domains, including autonomous driving, medical image analysis, satellite remote sensing, intelligent surveillance, and smart agriculture. In these areas, computer vision systems support digital transformation by enabling more accurate perception, efficient analysis, and informed decision-making (Chen et al., 2025; Xu, Wang, Jiang, & Gong, 2024).

Machine learning models also benefit from their ability to improve as they are exposed to increasing amounts of data, which is particularly important for image and video restoration tasks that require learning fine visual details. Bergmann (2025) emphasized that this data-driven improvement allows models to capture subtle patterns that are difficult to encode manually. Among image restoration problems, motion deblurring remains especially challenging because it requires recovering textures and structures lost due to complex camera or object motion in real-world conditions (Zhang et al., 2022).

Supervised learning has become one of the most commonly adopted approaches for motion deblurring. This paradigm relies on paired blurred and sharp images, allowing models to learn a direct mapping from degraded inputs to clear outputs. In this setting, motion deblurring is treated as an image-to-image translation problem. By comparing blurred images with their corresponding ground-truth sharp versions during training, deep learning models learn detailed relationships that enable the recovery of fine textures and high-frequency image information (Zhang et al., 2022; Ho, Duong, Lee, & Hong, 2025).

Prior to the widespread adoption of deep learning, motion deblurring was primarily addressed using traditional image processing techniques. These approaches focused on estimating blur kernels or applying handcrafted priors to model the blur process explicitly. Common methods include Richardson–Lucy deconvolution, Wiener filtering, and Markov random field–based techniques (Huihui et al., 2023). The effectiveness of these methods strongly depends on accurate blur kernel estimation. However, in real-world scenarios, motion blur is often non-uniform and difficult to parameterize, which limits the performance of traditional approaches (Xiang et al., 2024). Additionally, these methods typically involve complex computations, are sensitive to noise, require heuristic parameter tuning, and exhibit limited generalization across different scenes (Jiang & Liu, 2022).

Deep learning–based methods address many of these limitations by eliminating the need for explicit blur kernel modeling. Instead, neural networks learn the deblurring process end to end from paired training data, allowing them to adapt to diverse blur patterns and complex imaging conditions. As a result, deep learning models generally achieve better restoration quality, improved robustness, and stronger generalization compared to traditional techniques (Xiang et al., 2024). This shift from handcrafted models to data-driven learning represents a significant development in motion deblurring research.

Blind and Non-Blind Image Deblurring

Image deblurring aims to recover a latent sharp image from a blurred observation. Based on whether the blur kernel PSF is known, deblurring methods are commonly classified into non-blind and blind approaches.

According to Zhang et al. (2022), non-blind deblurring assumes that the blur kernel is given in advance. Even under this assumption, restoration remains challenging due to sensor noise and the loss of high-frequency image details. Early non-blind methods relied on natural image priors in the spatial or frequency domain, while more recent approaches combine traditional deconvolution with deep neural networks to reduce ringing artifacts, handle saturation, and suppress noise. Existing deep learning–based non-blind methods can be broadly grouped into approaches that perform deconvolution followed by denoising and those that directly employ deep networks for image restoration.

In contrast, blind deblurring does not assume prior knowledge of the blur kernel. Jiang et al. (2024) emphasized that blind deblurring is more difficult because both the latent sharp image and the blur kernel must be estimated simultaneously, making the problem highly ill-posed and sensitive to modeling errors. Traditional model-based blind deblurring techniques often struggle under real-world conditions due to unknown and highly variable degradation factors. Consequently, deep learning–based blind image restoration methods have become dominant, as they can learn complex degradation patterns directly from data and perform end-to-end restoration without explicit kernel estimation (Zhai et al., 2023). This significantly improved blind deblurring performance, particularly for complex and unknown motion blur scenarios.

Deep Learning Architectures for Blind Motion Deblurring

Deep learning has become central to advancing blind motion deblurring, as it enables models to learn complex mappings between blurred and sharp images. This section presents major deep learning approaches used for blind motion deblurring, focusing on Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), Recurrent Neural Networks (RNNs), and Transformer-based models.

Convolutional Neural Network-Based Models. CNNs are a category of machine learning models designed for visual data analysis and are widely applied to image and video tasks (Gillis, Craig, & Awati, 2024). CNN-based deblurring methods generally fall into two architectures: early two-stage frameworks and modern end-to-end approaches.

Early two-stage models first estimate a blur kernel using a neural network and then apply deconvolution or inverse filtering to reconstruct the sharp image. In contrast, end-to-end models generate a clear image directly from the blurred input. This approach allows the network to learn complex feature mappings without explicit kernel estimation (Xiang et al., 2024).

CNNs effectively capture local spatial features and are suitable for image denoising and deblurring tasks (Liu, Pu, & Sun, 2021; Tian et al., 2020). They adapt well to different blur types and perform efficiently when trained on large datasets. However, CNNs rely heavily on extensive paired training data and are limited by fixed receptive fields, which restrict their ability to model long-range dependencies. Dilated convolutions are commonly employed to address this limitation by enlarging the receptive field without increasing computational complexity (Zheng, Li, Zhu, Zhao & Huo, 2025; Niu, Han, & Cai, 2025).

Generative Adversarial Network-Based Models. GANs consist of a generator and a discriminator trained adversarially, where the generator synthesizes outputs and the discriminator distinguishes between real and generated data (AWS, n.d.). In deblurring, the generator directly takes blurred images as input to produce restored images intended to resemble clear images. At the same time, the discriminator receives both real sharp images and generated outputs to distinguish between real and restored images, thereby guiding the generator toward more realistic deblurring results (Li, 2024).

GANs provide advantages in enhancing perceptual realism and handling diverse forms of blur, which improves generalization across different scenarios. Since discriminator feedback drives learning, GAN-based models can utilize unlabeled data and operate in supervised or unsupervised settings. Their game-theoretic training approach enhances parameter fitting accuracy. However, GAN training is prone to instability, imbalance between the generator and discriminator, and issues such as mode collapse or non-convergence (Cui, Sun, Kong, Zhang, & Li, 2020).

Recurrent Neural Network-Based Models. RNNs generate sequential outputs by incorporating both current input and previous states, making them effective for sequential or time-dependent data (Singh, 2022). Although originally developed for text, language, and time-series applications, RNNs have been adopted in image processing due to their memory-based architecture (Yin, Yin, Chen, Huang & Liu, 2024).

In deblurring tasks, RNNs are advantageous for modeling temporal information, particularly in motion blur scenarios where blur patterns evolve over time. However, RNNs struggle to capture spatial information and often face vanishing or exploding gradient problems when modeling long-term dependencies (Rusch & Mishra, 2021; Ren et al., 2021). For this reason, RNNs are typically combined with other architectures to enhance performance in image deblurring tasks.

Transformer-Based Models. Transformers were initially developed for sequence transduction tasks in neural machine translation and operate by learning contextual relationships within sequential data (Ferrer, 2024). By using self-attention mechanisms, Transformers capture long-range dependencies and adaptively assign importance to different regions of an input (Liu et al., 2021).

In motion deblurring, Transformers provide significant improvements by capturing global information and modeling remote spatial dependencies (Yan et al., 2023). Their ability to process multi-scale features enables flexible adaptation to diverse blur patterns (Xiang et al., 2024). Positional encoding further strengthens their capacity to understand spatial relationships within images, while dynamic attention allocation helps address varying blur severity across regions of an image (Zhao et al., 2023).

Evaluation Metrics in Motion Deblurring

Image Quality Assessment (IQA) plays a crucial role in evaluating the effectiveness of image restoration methods, including motion deblurring. IQA is a fundamental challenge in image-based technologies and strongly influences the progress of image processing and computer vision research, as it provides standardized benchmarks for assessing algorithm performance (Ma et al., 2025). IQA methods are commonly used to assess the performance of image processing algorithms, including dehazing, medical image enhancement, and image deblurring, by quantifying how closely the processed image corresponds to a reference image. These metrics may target specific distortions, such as blurring, blocking, or ringing, or provide an overall measure of image fidelity across multiple degradation types (Sara, Akter, & Uddin, 2019).

Over time, IQA techniques have evolved from simple statistical measures to traditional machine learning approaches, such as support vector regression–based models, and more recently to deep learning–based methods including convolutional neural networks, Transformer-based architectures, and advanced training frameworks. Despite these developments, motion deblurring research continues to rely predominantly on traditional full-reference metrics, particularly Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index Measure (SSIM), due to their simplicity, interpretability, and widespread acceptance in benchmarking studies (Li, 2022).

Mean Squared Error. Mean Squared Error (MSE) measures the average squared difference between the restored image and the ground-truth sharp image. It provides a direct pixel-wise error measurement, where lower values indicate smaller discrepancies between the restored and reference images and, consequently, better restoration quality (Li, 2022). MSE is defined as:

$$\mathrm{MSE}(I, S) = \frac{1}{K} \sum_{k=0}^{K-1} (I_k - S_k)^2$$

where I represents the restored image, S represents the ground-truth sharp image, and K denotes the total number of pixels in the image.

Peak Signal-to-Noise Ratio. PSNR is a widely used metric for evaluating image deblurring performance by expressing reconstruction accuracy in terms of pixel-level differences. PSNR is derived from MSE and is defined as:

$$\mathrm{PSNR}(I, S) = 10 \log_{10} \left(\frac{(2^n - 1)^2}{\mathrm{MSE}(I, S)} \right)$$

where n denotes the number of bits used to represent each pixel value. Higher PSNR values indicate that the restored image is closer to the ground-truth image. However, PSNR primarily reflects numerical fidelity and does not account for perceptual quality, which limits its effectiveness when used alone. For this reason, PSNR is commonly combined with other metrics for comprehensive evaluation (Hore & Ziou, 2010).

Structural Similarity Index Measure. The SSIM evaluates image quality by comparing structural information between the restored image and the reference image. Unlike pixel-based metrics, SSIM considers perceptual factors and aligns more closely with human visual perception. SSIM assesses image similarity based on three components: luminance, contrast, and structure (Wang et al., 2004). SSIM is defined as:

$$\mathrm{SSIM}(I,S) = l(I,S)^{\alpha}\, c(I,S)^{\beta}\, s(I,S)^{\gamma}$$

where $l(I,S)$ represents the brightness comparison of the image, $c(I,S)$ represents the contrast comparison of the image, $s(I,S)$ represents the structural comparison of the image. In practical applications, it is common to set $\alpha = \beta = \gamma = 1$. Each component is computed as:

$$l(I,S) = \frac{2\mu_I\mu_S + C_1}{\mu_I^2 + \mu_S^2 + C_1}, \quad c(I,S) = \frac{2\sigma_I\sigma_S + C_2}{\sigma_I^2 + \sigma_S^2 + C_2}, \quad s(I,S) = \frac{\sigma_{IS} + C_3}{\sigma_I\sigma_S + C_3}$$

where μI and μS represent the mean intensities, σI and σS denote the standard deviations, and σIS is the covariance between images I and S, defined as:

$$\sigma_{IS} = \frac{1}{K}\sum_{k=0}^{K-1}(I_k - \mu_I)(S_k - \mu_S)$$

Constants C_1, C_2, and C_3 are included to avoid numerical instability. Higher SSIM values indicate greater structural similarity between the restored and reference images, signifying better perceptual restoration quality.

PSNR and SSIM are selected as evaluation metrics because they provide complementary and sufficient measures of motion deblurring performance. PSNR quantifies pixel-level reconstruction accuracy, while SSIM evaluates structural similarity in a manner consistent with human visual perception. Together, these metrics adequately capture both numerical fidelity and perceptual quality, making them suitable for assessing restoration performance and for comparison with existing motion deblurring studies.

Overview of Statistical Analysis in Research

Statistical analysis plays a critical role in research that aims to produce quantitative conclusions. It provides systematic methods for organizing, summarizing, and interpreting data. According to Vandever (2020), statistical analysis serves as a foundation for understanding research results and making valid inferences. The author emphasized that statistical testing should be carefully selected based on research objectives and data characteristics. Statistical software such as R, SAS, and SPSS are commonly used to perform analyses efficiently and minimize computational errors.

Statistical analysis is particularly important in retrospective and experimental research designs where data interpretation determines the validity of conclusions. It allows researchers to evaluate hypotheses objectively and determine whether observed differences are statistically significant.

Population Considerations and Statistical Errors. One of the fundamental considerations in statistical analysis is defining the appropriate population. Vandever (2020) explained that research findings are only applicable to the population represented by the data. In experimental and benchmarking studies, conclusions are limited to the dataset used for evaluation.

Two types of statistical errors are commonly discussed in hypothesis testing. A Type I error occurs when the null hypothesis is rejected even though it is true. This error is represented by alpha (α) and is also known as a false positive. A Type II error occurs when the null hypothesis is not rejected even though it is false. This is represented by beta (β) and is also known as a false negative. Careful selection of the significance level is necessary to balance these risks during statistical analysis (Vandever, 2020).

Parametric Tests for Comparing Means. Several statistical methods are used to test differences between group means. Mishra, Singh, Pandey, Mishra, and

Pandey (2019) discussed the use of the Student's t test, Analysis of Variance (ANOVA), and Analysis of Covariance (ANCOVA) in hypothesis testing.

The Student's t test is used to compare the means of two independent groups. It determines whether the difference between group averages is statistically significant. When more than two groups are compared, ANOVA is commonly applied. ANOVA produces a single p-value that indicates whether at least one group differs significantly from the others. If the result is significant, post hoc multiple comparison tests are conducted to identify the specific group differences.

ANOVA can be classified into one-way or two-way designs. One-way ANOVA involves one categorical independent variable, while two-way ANOVA involves two categorical independent variables. When a covariate is included to control for additional variability, the method becomes ANCOVA.

Mishra et al. (2019) also noted that parametric tests assume normal distribution of the dependent variable. Small sample sizes may increase sensitivity to outliers, which can distort mean values and affect test results. Therefore, sample adequacy and assumption checking are essential before applying these methods.

Two-Way Analysis of Variance. Two-Way ANOVA expands the application of one-way ANOVA by incorporating two independent variables instead of one. While one-way ANOVA examines the effect of a single factor on a dependent variable, two-way ANOVA analyzes the influence of two categorical independent variables on one continuous dependent variable. According to Mishra et al. (2019), this method is used to determine both the individual effects of each independent variable and the possible interaction between them.

In two-way ANOVA, the dependent variable is required to be continuous and approximately normally distributed. The two independent variables must be categorical. The analysis evaluates three components: the main effect of the first factor, the main effect of the second factor, and the interaction effect. The interaction effect indicates whether the impact of one independent variable varies depending on the level of the other variable.

Mishra et al. (2019) also stated that two-way ANOVA can be performed using statistical software such as SPSS under the General Linear Model through the Analyze – General Linear Model – Univariate option.

Statistical Method to Be Employed in the Proposed Study. Since the study evaluates the effects of two independent variables, namely hardware stabilization condition (with OIS and without OIS) and model adaptation condition (pre-trained and fine-tuned), on continuous performance metrics (PSNR and SSIM), a Two-Way ANOVA will be employed.

This method will allow the analysis of (1) the main effect of hardware stabilization, (2) the main effect of domain-adaptive fine-tuning, and (3) the interaction effect between stabilization condition and model adaptation.

The simplified two-way ANOVA model is expressed as:

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

where:

Y_{ij} : Observed performance metric (PSNR or SSIM)
μ : Overall mean
α_i : Effect of hardware stabilization condition
β_j : Effect of model adaptation condition
$(\alpha\beta)_{ij}$: Interaction effect between the two factors
ε_{ijk} : Random error term

In this study, Factor A will represent the hardware stabilization condition, consisting of images captured with OIS and without OIS. Factor B will represent the model adaptation condition, consisting of pre-trained and fine-tuned models. These two factors will correspond to the main effects and interaction term to be evaluated in the statistical model.

The effects of hardware stabilization, model adaptation, and their interaction will be tested using the following F-statistics:

$$F_A = \frac{MS_A}{MS_E}, \quad F_B = \frac{MS_B}{MS_E}, \quad F_{AB} = \frac{MS_{AB}}{MS_E}$$

where F_A will test the main effect of hardware stabilization, F_B will test the main effect of model adaptation, and F_{AB} will test the interaction effect between the two factors.

Performance Evaluation and Measurement of Change

Performance evaluation in experimental and computational studies often involves measuring the change between an initial and a final value. Two commonly used metrics for quantifying change are absolute change and relative change (Curran-Everett & Williams, 2015).

Absolute change. Absolute change refers to the direct difference between two measured values. It is computed by subtracting the initial value from the final value. Curran-Everett and Williams (2015) described absolute change as the simple numerical difference between two indicators measured across different periods. When indicators are expressed in percentage form, absolute change may also represent the difference in percentage points.

Mathematically, absolute change is expressed as:

$$\Delta y = y_f - y_i$$

where is the initial value and is the final value.

Relative change. Relative change expresses the magnitude of improvement relative to the initial value. It standardizes the change by dividing the absolute difference by the baseline measurement. Curran-Everett and Williams (2015) explained that relative change allows better comparison across different scales because it reflects proportional growth.

Mathematically, percent change is expressed as:

$$\%\Delta = \frac{y_f - y_i}{y_i} \times 100$$

This formulation converts the improvement into percentage form, making it easier to interpret the relative magnitude of change.

However, percent change has limitations. Curran-Everett and Williams (2015) noted that when the initial value is zero, percent change becomes undefined. In addition, as the initial value approaches zero, the percent change increases without bound. It may approach positive infinity when the final value is greater than the initial value, or negative infinity when the final value decreases.

These characteristics indicate that percent change can produce exaggerated results when baseline values are very small. Therefore, its interpretation in experimental studies requires caution, especially when improvement metrics depend on low initial measurements.

Effect Size. Effect size is a fundamental concept in empirical research because it indicates the practical importance of study findings. Lakens (2013) explained that effect sizes support cumulative research by allowing results to be compared across different studies. They also assist researchers in estimating the required sample size for future investigations and in assessing the consistency of findings across multiple experiments.

Eta Squared in Between- and Within-Subjects Comparisons. Eta squared (η^2) is an effect size measure that represents the proportion of total variance in a dependent variable that is attributed to an independent variable. It is calculated by dividing the sum of squares of the effect by the total sum of squares:

$$\eta^2 = \frac{SS_{\text{effect}}}{SS_{\text{total}}}$$

Although eta squared is useful within a single study, its value is influenced by the total variance in the design, particularly in factorial and repeated measures designs. Keppel (1991, as cited in Lakens, 2013) noted that as more variables are introduced, the total variance increases, which can affect the comparability of η^2 across different studies.

To address this limitation, researchers often report partial eta squared (ηp^2). Partial eta squared measures the proportion of variance explained by an effect relative to the variance associated with that effect and its corresponding error term (Lakens, 2013). It is computed as:

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{error}}}$$

Cohen (1988, as cited in Lakens, 2013) proposed general guidelines for interpreting effect sizes. Values of 0.01 indicate a small effect, 0.06 indicate a medium effect, and 0.14 indicate a large effect. These benchmarks help determine whether statistically significant results reflect practically meaningful differences, particularly in studies with large sample sizes where small effects may still reach significance.

Overall, the measurement of performance improvement in experimental studies requires both statistical and practical evaluation. Absolute change and percent change provide direct and proportional measures of improvement, allowing clearer comparison between baseline and adapted conditions. However, statistical significance alone is not sufficient to determine the meaningfulness of observed differences. Effect size measures, particularly partial eta squared, offer a standardized way to assess the magnitude of an effect within factorial designs. By combining change-based metrics with effect size interpretation, researchers can ensure that reported improvements are not only statistically detectable but also practically relevant. This integrated approach strengthens the validity and interpretability of performance evaluation in computational experiments.

Review of Related Literature

This section reviews existing literature relevant to motion deblurring, including synthetic and real-world blur modeling, OIS, deep learning architectures, dataset construction strategies, post-processing pipelines, and hyperparameter selection. The discussion emphasizes methodological approaches, benchmarking practices, and performance evaluation using PSNR and SSIM.

Synthetic and Real-World Motion Blur in Image Deblurring Benchmarks

This section reviews literature that compare synthetic and real-world motion blur in image deblurring benchmarks. While synthetic datasets enable large-scale training and evaluation, they often fail to reflect real-world blur characteristics, resulting in limited generalization and degraded performance on real images. Existing literature emphasizes the importance of blur realism and diversity in benchmarking deblurring models for practical deployment.

Synthetic Motion Blur in Learning-Based Deblurring. Ho et al. (2025) investigated supervised deep-learning-based motion deblurring models trained on synthetic blur datasets and reported that these models often suffer from overfitting to their training data. Although synthetic blur datasets are essential for providing large-scale paired sharp and blurred images, the trained models frequently fail to generalize to unseen blur domains in real-world scenarios. Their study showed that this limitation commonly results in undesired visual artifacts and degraded restoration quality when applied to real images. These findings suggest that synthetic blur alone is insufficient for developing robust and generalizable motion deblurring models.

Limitations of Synthetic Blur in Modeling Real-World Image Formation. Zhang et al. (2020) emphasized that synthetically generated blur does not accurately represent the physical image formation process observed in real-world motion blur. They explained that blur caused by camera motion and scene dynamics cannot be sufficiently modeled using conventional synthetic blurring techniques. To address this issue, they proposed a learning-based framework that first learns realistic blur generation from real blurred images before performing deblurring. Their results demonstrated that reducing the discrepancy between synthetic and real blur leads to improved perceptual quality and more reliable restoration performance.

Generalization Issues Between Synthetic and Real-World Motion Blur. Xu et al. (2021) examined the performance of motion deblurring models trained on synthetic data and evaluated their behavior on real-world motion blur. Their study showed that inconsistencies between synthetic and real data distributions cause significant performance degradation when models are applied to real-world scenarios. They further demonstrated that learning directly from real-world motion-blurred images enables more adaptive and practical deblurring performance. These findings highlight the importance of real-world data in improving generalization beyond synthetic training conditions.

Dataset Constraints and Blur Diversity in Benchmarking. Gao et al. (2026) analyzed existing motion deblurring benchmarks and identified dataset limitations as a major factor restricting real-world generalization. They reported that synthetic datasets provide large-scale coverage but diverge from real-world blur distributions, while real-captured datasets improve realism but remain limited in blur diversity and scene coverage. Their study identified blur pattern diversity as a decisive factor for robust generalization and proposed transferring blur priors from simulation datasets to real data through joint fine-tuning. Their results confirmed that insufficient blur diversity in existing benchmarks limits the ability of deblurring models to perform reliably under real-world conditions.

Optical Image Stabilization and Motion Blur

Motion blur commonly occurs in handheld image capture due to unintended camera movement during exposure. OIS is a hardware-based solution that reduces this effect by detecting camera motion through gyroscope sensors and compensating for it using lens element adjustments. Since this correction is applied before light reaches the image sensor, images captured with OIS often exhibit reduced blur and altered motion patterns compared to images captured without OIS. While this improves visual quality, it also changes the underlying motion characteristics recorded during image capture, which has implications for dataset construction and model evaluation (Liu et al., 2021).

Dataset Construction Using Sensor-Aided Approaches. Several studies have constructed datasets that incorporate gyroscope or Inertial Measurement Unit (IMU) data to support stabilization and deblurring tasks. Shi, Shi, Lai, Liang, and Liang (2022) introduced a dataset containing synchronized video frames, gyroscope readings, and OIS-related motion information for video stabilization. Their dataset enabled learning-based fusion of sensor data and image content. However, the dataset was designed for stabilization tasks and did not explicitly separate or compare samples captured with OIS and samples captured without OIS for image deblurring analysis.

Luan et al. (2024) proposed a framework for generating synthetic motion blur datasets using IMU data to produce pixel-aligned training triplets. Their dataset construction strategy closely approximates real-world motion blur and supports effective deblurring model training. Despite this, the dataset assumes a unified camera motion model and does not distinguish between images captured with OIS and images captured without OIS, which limits its suitability for comparative evaluation across different capture conditions.

OIS-Aware Data and Motion Compensation Datasets. The interaction between OIS and gyroscope-based image processing has been examined by Liu et al. (2021). They constructed a dataset using images and sensor data captured with OIS to study how stabilization affects gyroscope-based alignment. Their work focuses on compensating OIS-induced motion so that gyroscope data can be reliably used for image alignment. Although the dataset demonstrates that OIS effects can be learned and corrected, it is primarily designed for alignment tasks and does not provide paired datasets for evaluating motion deblurring performance on images captured with OIS and images captured without OIS.

In video deblurring research, Xu, Zhang, Fan, Wu, & Zuo (2025) introduced datasets that include both synthetic and real-world handheld videos. Their synthetic dataset simulates handheld motion with OIS, while the real-world dataset is captured using smartphones with OIS. Although these datasets improve realism, they do not include a controlled counterpart captured without OIS, making it difficult to isolate and analyze the specific impact of OIS on deblurring model performance.

Existing literature shows that dataset construction for motion stabilization and deblurring increasingly relies on sensor data and realistic motion modeling. However, current datasets are typically designed for a single capture condition, most commonly images captured with OIS, or they focus on compensating OIS effects rather than analyzing them. There is limited availability of datasets that are explicitly constructed to include both images captured with OIS and images captured without OIS under comparable capture conditions.

As a result, there remains insufficient data and limited comparative analysis examining how motion deblurring models perform when trained and evaluated on datasets composed of images captured with OIS and images captured without OIS. This gap highlights the need for a dedicated dataset construction strategy that enables direct comparison between these two capture conditions. Addressing this gap would support more reliable benchmarking and provide clearer insights into the robustness and generalization of motion deblurring models across different hardware stabilization settings.

Motion Deblurring Architectures

Performance Comparison of SOTA Motion Deblurring Methods. Table 1 presents a quantitative comparison of state-of-the-art image deblurring methods evaluated on the GoPro, HIDE, RealBlur-J, and RealBlur-R datasets, as reported by Xiang et al. (2024). Performance is measured using PSNR and SSIM, which reflect pixel-level accuracy and perceptual structural consistency, respectively.

The results indicate that Transformer-based models consistently achieve high PSNR and SSIM values across datasets. FFTformer records the highest PSNR of 34.21 dB and an SSIM of 0.969 on the GoPro dataset. Other Transformer models, such as Stoformer and Uformer, also report PSNR values above 32.9 dB with SSIM values exceeding 0.96, placing them in the upper tier of performance. These results suggest strong capability in restoring both pixel accuracy and structural details.

CNN-based models demonstrate competitive performance in both metrics. Methods such as LaKDNet and MSSNet achieve PSNR values above 33 dB and SSIM values around 0.96 on the GoPro dataset. Despite this strong performance, these models do not exceed the highest PSNR or SSIM values achieved by Transformer-based approaches. RNN-based methods show moderate results, with RNN-MBP achieving a PSNR of 33.32 dB and an SSIM of 0.963, but overall performance remains slightly lower than that of Transformers. In contrast, GAN-based models consistently report lower PSNR and SSIM values. For example, CycleGAN records a PSNR of 22.54 dB and an SSIM of 0.720 on the GoPro dataset, indicating weaker pixel fidelity and structural preservation.

Table 1. Motion Deblurring Methods Performance Comparison (Xiang et al., 2024)

Methods	Category	GoPro (PSNR/SSIM)	HIDE (PSNR/SSIM)	RealBlur-J (PSNR/SSIM)	Realblur-R (PSNR/SSIM)
DeepDeblur	CNN	29.08 / 0.914	25.73 / 0.874	27.87 / 0.834	32.51 / 0.841
DMPHN	CNN	31.20 / 0.940	29.09 / 0.924	28.42 / 0.860	35.70 / 0.948
SDWNet	CNN	31.26 / 0.966	28.99 / 0.957	28.61 / 0.867	35.85 / 0.948
MSCAN	CNN	31.24 / 0.945	29.63 / 0.920	-	-
PSS-NSC	CNN	31.58 / 0.948	-	-	-
MIMOU-Net+	CNN	32.45 / 0.957	29.99 / 0.930	27.63 / 0.837	35.54 / 0.947
MPRNet	CNN	32.66 / 0.959	30.96 / 0.939	28.70 / 0.873	35.99 / 0.952
HINet	CNN	32.77 / 0.959	-	-	-
BANet	CNN	32.54 / 0.957	30.16 / 0.930	-	-
SAPHNet	CNN	32.02 / 0.953	29.98 / 0.930	-	-
MSSNet	CNN	33.01 / 0.961	30.79 / 0.939	28.79 / 0.879	35.93 / 0.953
MSFS-Net	CNN	32.73 / 0.959	31.05 / 0.941	28.97 / 0.908	36.02 / 0.959
LaKDNet	CNN	33.35 / 0.964	31.21 / 0.943	28.78 / 0.878	35.91 / 0.954
MRLPFNet	CNN	33.50 / 0.965	31.63 / 0.946	-	-
SRN	RNN	30.26 / 0.934	28.36 / 0.915	28.56 / 0.867	35.66 / 0.947
MT-RNN	RNN	31.15 / 0.945	29.15 / 0.918	28.44 / 0.862	35.79 / 0.951
ESTRNN	RNN	31.07 / 0.902	-	-	-
IFI-RNN	RNN	29.97 / 0.895	-	-	-
RNN-MBP	RNN	33.32 / 0.963	-	-	-
DeblurGAN	GAN	28.70 / 0.858	24.51 / 0.871	27.97 / 0.834	33.79 / 0.903
DeblurGAN-V2	GAN	29.55 / 0.934	26.61 / 0.875	28.70 / 0.866	35.26 / 0.944
DBGAN	GAN	31.10 / 0.942	28.94 / 0.915	24.93 / 0.745	33.78 / 0.909
CycleGAN	GAN	22.54 / 0.720	21.81 / 0.690	19.79 / 0.633	12.38 / 0.242
FCL-GAN	GAN	24.84 / 0.771	23.43 / 0.732	25.35 / 0.736	28.37 / 0.663
Ghost-GAN	GAN	28.75 / 0.919	-	-	-
CTMS	Transformer	32.73 / 0.959	31.05 / 0.940	27.18 / 0.883	-
Uformer	Transformer	32.97 / 0.967	30.83 / 0.952	29.06 / 0.884	36.22 / 0.957
Restormer	Transformer	32.92 / 0.961	31.22 / 0.942	28.96 / 0.879	36.19 / 0.957
Stripformer	Transformer	33.08 / 0.962	31.03 / 0.940	-	-
Stoformer	Transformer	33.24 / 0.964	30.99 / 0.941	-	-
FFTformer	Transformer	34.21 / 0.969	31.62 / 0.945	-	-
Sharpformer	Transformer	33.10 / 0.963	-	-	-

Mean Performance Analysis by Model Category. Table 2 summarizes the mean PSNR and SSIM values of the four architectural categories across the GoPro, HIDE, RealBlur-J, and RealBlur-R datasets. Transformer-based models achieve the highest mean PSNR and SSIM in seven out of eight evaluated metrics, demonstrating consistent performance across different datasets.

On the GoPro dataset, Transformer-based methods record a mean PSNR of 33.179 dB and a mean SSIM of 0.964, outperforming CNN, RNN, and GAN architectures. This trend is also observed on the HIDE dataset, where Transformers achieve the highest mean SSIM of 0.943. The advantage of Transformer-based models is most pronounced on the RealBlur-R dataset, with a mean PSNR of 36.205 dB and a mean SSIM of 0.957, indicating strong structural similarity preservation under real-world blur conditions.

CNN-based models consistently rank second in both PSNR and SSIM across most datasets and demonstrate a marginal advantage in mean PSNR on the RealBlur-J dataset. RNN-based models follow closely but show slightly lower mean SSIM values compared to CNN and Transformer approaches. GAN-based models record the lowest mean PSNR and SSIM across all datasets, suggesting limitations in maintaining both pixel-level accuracy and structural consistency during image restoration.

Table 2. Mean Performance by Model Category

Category	GoPro (Mean PSNR)	GoPro (Mean SSIM)	HIDE (Mean PSNR)	HIDE (Mean SSIM)	RealBlur -J (Mean PSNR)	RealBlur -J (Mean SSIM)	Realblur -R (Mean PSNR)	Realblur -R (Mean SSIM)
Transformer	**33.179**	**0.964**	**31.123**	**0.943**	28.400	**0.882**	**36.205**	**0.957**
CNN	32.099	0.953	29.934	0.931	**28.471**	0.867	35.431	0.938
RNN	31.154	0.928	28.755	0.917	28.500	0.864	35.725	0.949
GAN	27.580	0.857	25.060	0.817	25.348	0.763	28.716	0.732

Justification for Using a Transformer-Based Model. The results presented in Tables 1 and 2 indicate that Transformer-based models provide the most consistent and balanced performance in image deblurring tasks when evaluated using both PSNR and SSIM. Their ability to achieve high PSNR values reflects strong pixel-level restoration, while consistently high SSIM values indicate effective preservation of structural information and visual quality.

The superior performance of Transformer-based models can be attributed to their self-attention mechanism, which enables modeling of long-range dependencies and global spatial relationships within blurred images. This capability is particularly important for motion blur, where distortions often extend across large image regions and affect overall structure.

Although CNN and RNN architectures demonstrate competitive PSNR and SSIM values, their performance is generally surpassed by Transformer-based approaches across multiple datasets. GAN-based models, while capable of generating visually plausible textures, consistently exhibit lower PSNR and SSIM values, making them less suitable for applications that prioritize accurate structural and pixel-level restoration. Based on these findings, the use of a Transformer-based model is well justified for the proposed image deblurring approach in this study.

Transformer-Based Models for Motion Deblurring

Recent advances in motion deblurring have shown that Transformer-based architectures provide strong performance in modeling complex motion blur. Several studies have proposed Transformer-based models that improve image restoration by capturing long-range dependencies, global contextual information, and multi-scale features. However, existing approaches differ in architectural design, computational efficiency, and their ability to preserve fine details. This section reviews representative Transformer-based image deblurring models and compares their strengths and limitations to establish the motivation for selecting MLWNet as the pretrained model for this study.

Comparison of Transformer-Based Motion Deblurring Models. Table 3 presents a comparison of representative Transformer-based image deblurring methods reported in recent literature. The comparison highlights their publication venue, core architectural ideas, advantages, and known limitations. This analysis provides the basis for selecting an appropriate pretrained model for the proposed study.

Table 3. Comparison of Transformer-Based Motion Deblurring Models

Methods	Pub.	Overview	Advantages	Limitations
Uformer	CVPR 2021	Builds a hierarchical encoder–decoder architecture using Transformer blocks to capture multi-scale features	Captures both local and global dependencies effectively	Self-attention layers can only be applied to limited depth due to high computational cost
Restormer	CVPR 2022	Learns multi-scale local–global representations using efficient Transformer designs for high-resolution images	Models global connectivity with high computational efficiency	Spatial information is not fully exploited during image recovery
Stripformer	ECCV 2022	Re-weights image features horizontally and vertically to handle directional motion blur	Low memory footprint and computational cost; less reliance on large training datasets	Scaled dot-product attention involves complex matrix multiplication
FFTformer	CVPR 2023	Combines self-attention with frequency-domain modeling using FFT-based operations	Lower space and time complexity; effective blur removal	May not generalize well across diverse blur types and image domains
Sharpformer	TIP 2023	Uses dynamic convolution guided by global–local features for non-uniform blur	Improves deblurring quality compared to CNN-only approaches	High computational complexity
MLWNet	**CVPR 2024**	Proposes a single-input multiple-output multi-scale Transformer-based network with a learnable discrete wavelet transform for motion deblurring	Efficient multi-scale restoration; strong detail recovery using learnable wavelet-based frequency modeling; high performance on real-world blur	Lower performance on synthetic blur due to sensitivity to unnatural frequency distributions

Justification for Using MLWNet as a Pretrained Model. Table 3 shows that recent Transformer-based motion deblurring models primarily aim to improve the modeling of global dependencies and multi-scale features. Early approaches such as Uformer and Restormer demonstrate the effectiveness of Transformer architectures in capturing long-range spatial relationships. However, their performance is often constrained by computational cost or incomplete utilization of spatial information (Wang et al., 2022; Zamir et al., 2022).

More recent models, including Stripformer and FFTformer, attempt to reduce complexity by introducing directional attention or frequency-domain processing. While these approaches improve efficiency, they may face challenges in generalization or require complex attention operations (Tsai, Peng, Lin, Tsai & Lin, 2022; Kong, Dong, Ge, Li & Pan, 2023). Models such as Sharpformer further extend Transformer-based designs to handle non-uniform blur but introduce additional computational overhead (Yan et al., 2023).

MLWNet differs from earlier approaches by addressing limitations of traditional coarse-to-fine and multi-scale architectures. Instead of relying on multiple low-resolution inputs and complex fusion modules, MLWNet adopts a single-input multiple-output strategy that progressively restores images at different scales. This design simplifies multi-scale processing while maintaining strong semantic consistency across scales.

A key distinction of MLWNet is the integration of a learnable discrete wavelet transform. By explicitly modeling frequency components and directional continuity, MLWNet improves the restoration of fine details that are often degraded in multi-scale Transformer architectures. The use of wavelet-based feature decomposition allows the model to better separate and reconstruct high- and low-frequency information, which is critical for real-world motion blur.

Although MLWNet shows reduced performance on synthetic blur datasets, this limitation is attributed to the unnatural frequency distributions present in synthetic data rather than architectural weakness. Overall, the balance between restoration quality, computational efficiency, and detail preservation makes MLWNet a suitable pretrained Transformer-based model for the proposed image deblurring study (Gao et al., 2023).

Dataset Construction Strategies for Benchmarking

The dataset creation process is widely recognized as a critical yet often undervalued component of machine learning research, particularly in benchmarking studies. Orr and Crawford (2024) emphasized that the quality, scope, and ethical grounding of datasets directly influence the reliability, fairness, and safety of machine learning systems. They framed dataset creation as a multi-phase process, noting that weaknesses or oversights at any stage may lead to biased outcomes and unintended harms. As benchmarking relies on consistent and comparable datasets, careful attention to each phase of dataset construction is necessary to ensure valid and meaningful evaluation of models.

Parameter and Population Definition. Orr and Crawford (2024) described dataset creation as beginning with parameter and population definition. This phase involves identifying the variables of interest and determining which data fall within or outside the scope of the dataset. Clear definition at this stage establishes what the dataset is intended to represent and guides all subsequent decisions in the dataset construction process.

Data Collection. Following parameter and population definition, the dataset creation process proceeds to data collection. According to Orr and Crawford (2024), data collection may involve reuse of existing data sources, web-based collection, or human annotation. These approaches are often shaped by dataset objectives as well as practical and institutional constraints. The literature highlights that data collection strategies influence the diversity, coverage, and representativeness of the resulting dataset.

Data Cleaning. Once collected, datasets often contain errors, duplicates, or irrelevant entries. Orr and Crawford (2024) identified dataset cleaning as a necessary phase for correcting inaccuracies and removing noise. Cleaning improves consistency and prepares the dataset for meaningful analysis while preventing the propagation of flawed data into later stages

Data Validation. Dataset validation is conducted to assess whether the data accurately represent real-world conditions and conform to the dataset's intended assumptions. According to Orr and Crawford (2024), validation may reveal quality issues that require revisiting earlier phases, making dataset creation an iterative rather than linear process.

Dataset Splitting. Dataset splitting is a fundamental step in data preparation for machine learning and statistical modeling. It involves dividing a dataset into separate subsets to ensure that model performance is evaluated using unseen data. This process supports fair training and reliable performance assessment, particularly in benchmarking studies where objective comparison between models is required (Orr & Crawford, 2024).

In machine learning practice, datasets are commonly divided into training and testing subsets. The training set is used to estimate model parameters, while the testing set evaluates predictive performance on new data (Joseph & Vakayil, 2021). This separation is necessary to prevent overfitting, a condition where a model performs well on training data but fails to generalize to unseen instances. By reserving a portion of the dataset for testing, researchers obtain a more accurate estimate of real-world performance.

For more complex modeling tasks, the dataset may be divided into three subsets: training, validation, and testing. The validation set is used for hyperparameter tuning and performance monitoring during model development, while the testing set remains reserved for final evaluation (GeeksforGeeks, 2023). This three-way split further reduces the risk of overfitting and supports systematic model optimization.

The appropriate proportion of each subset depends on the size of the dataset. Kumar (2020) recommended a 60:20:20 ratio for datasets containing between 100 and 1,000,000 samples. For datasets exceeding one million samples, smaller validation and testing portions such as 98:1:1 or 99:0.5:0.5 may be sufficient. These recommendations highlight the need to balance training adequacy with reliable evaluation.

In addition to split ratios, the sampling technique used for partitioning also affects model performance stability. Reitermanova (2010) identified several sampling methods, including Simple Random Sampling (SRS), stratified sampling, and DUPLEX. While SRS is widely applied due to its simplicity and efficiency, it may introduce higher bias and variance in high-dimensional datasets. Stratified sampling, although more computationally demanding, ensures proportional representation of classes and is often more appropriate for large and complex datasets.

Overall, the literature consistently recognizes dataset splitting as a critical methodological step in machine learning research. Proper partitioning improves model generalization, minimizes overfitting, and ensures fair and objective evaluation.

Dataset Circulation. Dataset circulation refers to the process of making datasets available for reuse through repositories, organizational platforms, or research publications. Circulation allows datasets to support further benchmarking and comparative studies beyond their original purpose. Orr and Crawford (2024) noted that once datasets are released, they may be reused, redistributed, or modified beyond the control of their creators. This highlights the importance of careful design, documentation, and quality control during dataset creation to support responsible reuse and mitigate potential unintended impacts.

Dataset Size and Data Efficiency in Image Deblurring

Dataset size plays an important role in training deep learning-based image deblurring models. Lai, Huang, Hu, Ahuja and Yang (2016) demonstrated that effective evaluation of image deblurring models does not always require large datasets. The study used 100 real blurred images collected from multiple sources and captured under varied camera devices and settings. Images were annotated using five semantic attributes, including man-made scenes, natural scenes, presence of people or faces, saturated regions, and text. Despite the limited dataset size, the study was able to reveal clear performance differences among deblurring algorithms through perceptual evaluation. The findings emphasized that dataset diversity and realism are more critical than dataset size alone.

Yin, Ma, Lin, & Zheng (2024) further supported the findings of Lai et al. (2016) by analyzing data efficiency in image restoration tasks. Their results showed that training a model using only 100 to 300 images achieved performance nearly equivalent to training with the full dataset, with only a minimal difference in PSNR. This indicates diminishing performance gains as dataset size increases beyond a certain point. The study suggests that carefully curated datasets can achieve comparable results while reducing computational cost and training requirements.

Image Acquisition Process for Real-World Motion Blur

Image acquisition refers to the physical and technical procedures used to capture images using camera devices. In motion deblurring research, this process is critical because blur is formed during exposure at capture time. The realism of a deblurring dataset depends on how accurately the acquisition setup reflects real-world imaging conditions.

Beam Splitter–Based Image Acquisition Using High-End Cameras. Many real-world deblurring datasets are collected using beam splitter systems combined with high-end imaging devices. Rim et al. (2020) introduced a dual-camera system using a beam splitter and two identical mirrorless cameras. One camera captured blurred images using a long shutter speed, while the other captured sharp images using a short shutter speed. The cameras were synchronized to ensure simultaneous exposure, allowing real motion blur to be formed physically during capture.

This acquisition strategy has been widely adopted. Zhong, Gao, Zheng, and Zheng (2020) and Zhong, Gao, Zheng, Zheng, & Sato (2021) collected paired blurry and sharp video clips using co-axis beam splitter systems with controlled exposure times and irradiance levels. Rim et al. (2022) further extended this approach by capturing multiple sharp images during the exposure of a blurred image, selecting the temporally centered sharp frame as ground truth. Li et al. (2023) focused on local motion blur by using synchronized scientific cameras and beam splitters, with exposure times adjusted according to object distance and motion speed.

While these systems produce well-aligned image pairs, they rely on professional mirrorless or scientific cameras with large sensors, interchangeable lenses, and controlled optics. Such imaging conditions differ significantly from those of consumer mobile devices.

Frame Averaging from High-Speed Cameras. Another widely used strategy for generating blurred and sharp image pairs is frame averaging from high-speed cameras. This approach is based on the observation that motion blur can be modeled as the temporal integration of sharp images during the camera's exposure time. Instead of capturing a blurred image directly, a sequence of sharp frames is recorded at a high frame rate and averaged to synthesize blur.

Nah et al. (2017) adopted this approach by capturing high-speed video sequences of dynamic scenes. A blurry image was generated by averaging multiple sharp frames acquired during the simulated shutter interval, with gamma correction applied to approximate the camera response function. This method enabled precise control over blur strength while maintaining temporal alignment between sharp and blurred images.

Carbajal, Vitoria, Lezama, and Muse (2022) described high-speed camera–based blur generation as a common practice in popular benchmarks such as GoPro, DVD, and REDS. In these datasets, short video sequences were captured using high-frame-rate cameras. The middle frame of each sequence was treated as the sharp ground truth, while the blurred image was synthesized by averaging all frames in the sequence. Gamma correction was often applied to approximate real camera response characteristics. This strategy provided a discrete approximation of the physical blur formation process.

Although frame averaging from high-speed cameras is widely used and effective for controlled blur synthesis, it does not involve capturing blur directly during exposure. As a result, the generated blur may not fully reflect the complex interactions between camera hardware, sensor behavior, and real-world motion.

High-Speed Frame Averaging Using Mobile Devices. More recent work explored frame averaging using consumer-grade mobile devices rather than specialized high-speed cameras. Mahmud et al. (2025) utilized the high-speed video recording capability of a modern smartphone to construct a deblurring dataset under real-world conditions. Video sequences were captured at 240 frames per second using a fixed hardware setup, and blurred images were synthesized by averaging temporally aligned sharp frames. The middle frame was used as the sharp reference.

The use of a smartphone enabled dense temporal sampling of real-world motion while maintaining imaging characteristics representative of consumer photography. Compared with professional camera systems, mobile devices reflect smaller sensor sizes, built-in stabilization mechanisms, and computational photography pipelines. This approach demonstrated that high-speed frame averaging can be performed using widely available mobile hardware, improving dataset accessibility and reproducibility.

Mobile Device–Based Image Acquisition Using a Side-by-Side Rig. In addition to frame averaging, some studies captured real blur directly using mobile devices. Nahli et al. (2024) mounted multiple smartphones side-by-side on a rigid rig and simultaneously recorded videos in dynamic scenes. The recordings were aligned, trimmed to the same duration, and sampled for comparison. By capturing the same scene at the same time, this setup ensured consistent motion and lighting conditions across devices.

This acquisition strategy is notable because it reflects how images are commonly captured in practice. Mobile devices rely on small sensors, fixed lenses, and integrated image signal processing pipelines, which strongly influence blur formation. The study demonstrated that real motion blur can be captured directly on smartphones without specialized camera systems.

Research Gap in Image Acquisition for Mobile-Based Deblurring. The reviewed literature shows that most real-world motion deblurring datasets rely on either beam splitter systems with professional cameras or blur synthesized through high-speed frame averaging. While these approaches produce well-aligned image pairs, they are based on imaging pipelines and capture conditions that differ significantly from those of consumer mobile devices.

Recent studies have explored mobile-based acquisition using high-speed frame averaging or side-by-side smartphone rigs. However, these approaches remain limited in their ability to generate physically controlled and repeatable motion blur during image capture. In particular, existing mobile-based setups often lack mechanisms to systematically control motion patterns and blur intensity while preserving real exposure-time blur formation.

This reveals a gap in image acquisition methods that use consumer smartphones and incorporate controlled physical motion to mimic common real-world blur characteristics, such as linear motion and vibration-induced camera shake, under practical imaging conditions.

Paired Image Post-processing Pipeline

Paired image post-processing pipelines are essential in image deblurring research, where sharp and blurred image pairs are required for training and evaluation. These pipelines aim to reduce inconsistencies caused by sensor noise, camera misalignment, and photometric differences. Without systematic post-processing, learning-based deblurring models may be influenced by dataset artifacts rather than true motion blur characteristics.

These challenges have been addressed in the literature through a set of distinct yet related post-processing stages, which are discussed in the following subsections.

Image Signal Processing Operations. Several studies incorporated standard image signal processing operations into their post-processing pipelines. Rim et al. (2020) applied white balancing, demosaicing, cropping, lens distortion correction, and downsampling prior to alignment. These steps minimized structural and photometric artifacts early in the pipeline.

Similarly, Li et al. (2023) performed demosaicing, white balancing, color mapping, and gamma correction after photometric alignment. Applying these operations in a consistent order ensured reliable RGB image generation from raw sensor data.

Blur Detection and Frame Quality Assessment. Nishanth, Kumar, Krithi, Shetty, and Sharma (2025) investigated blur detection using edge-based and frequency-based features, with particular emphasis on the Laplacian operator. Their study demonstrated that motion blur significantly reduces Laplacian edge responses, as blurred images exhibit weakened high-frequency content and diminished edge strength. The variance of the Laplacian was shown to be an effective indicator of image sharpness, where lower values correspond to increased blur.

The findings of Nishanth et al. (2025) support the use of Laplacian-based measures for objective frame quality assessment. In line with this, the present study will employ Laplacian-based sharpness measures to objectively distinguish sharp reference frames from motion-blurred frames during the dataset creation process. By using reduced Laplacian responses as an indicator of motion blur, the frame selection process will be made more consistent and less dependent on subjective visual inspection.

Paired Data Consistency and Frame Selection. Mahmud et al. (2025) emphasized the importance of temporal and photometric consistency when constructing paired sharp–blurred data from high–frame-rate smartphone videos. Although their work primarily investigated blur synthesis through frame averaging, it highlighted a broader issue that will remain relevant for real captured data: the need for consistent and reliable frame selection under controlled acquisition conditions. Prior studies suggest that identifying a valid sharp reference frame is a critical prerequisite for establishing accurate sharp–blur pairs, regardless of whether blur is synthetically generated or naturally captured. In this context, objective sharpness measures, such as Laplacian-based metrics, will be relevant for distinguishing sharp frames from motion-blurred observations within a video sequence. By applying such measures, future dataset construction efforts are expected to ensure that paired sharp and blurred frames originate from the same scene and share consistent photometric characteristics, with differences arising primarily from motion-induced blur rather than acquisition variability.

Geometric Alignment of Paired Images. Geometric misalignment commonly arises from mechanical inaccuracies and shutter timing differences. Rim et al. (2020) proposed a three-stage alignment approach consisting of homography-based alignment, phase correlation refinement, and centroid-based alignment using estimated blur kernels. Each stage addressed different sources of misalignment, resulting in improved pixel-level correspondence.

Li et al. (2023) employed optical flow estimation to geometrically align blurred images to their corresponding sharp images. Using static reference image pairs, a calibrated optical flow field was applied consistently across image pairs within the same scene. This approach maintained spatial consistency while reducing alignment errors, particularly along object boundaries.

Photometric and Color Alignment. Photometric differences between paired images can persist even when identical camera models are used. Rim et al. (2020)

addressed this issue through linear photometric alignment, adjusting sharp images to match the intensity statistics of blurred images on a per-channel basis. Estimating alignment parameters from reference image pairs reduced the effect of blur during correction.

Zhao, Ferguson, Zhou, Elliott, and Rafferty (2022) formulated color alignment as a calibration-based process that corrects sensor response differences and matches color distributions between images captured under varying conditions. Their approach enabled direct alignment of a target image to a reference image without requiring ground-truth color information. By normalizing color distributions, the method ensured consistent chromatic characteristics while preserving structural content. The study emphasized the role of color alignment as a critical preprocessing step for achieving relative color constancy.

Resolution Adjustment and Interpolation. Resolution adjustment is commonly applied in paired image post-processing pipelines to suppress sensor noise, reduce minor parallax errors, and decrease computational complexity. Downsampling reduces spatial resolution and removes high-frequency noise components, thereby improving structural consistency between sharp and blurred image pairs. Rim et al. (2020) applied spatial downsampling to sharp images captured under high ISO settings to minimize noise amplification while preserving essential structural details.

Image resizing requires interpolation to compute new pixel values in the resized image grid. Each pixel in the resized image is mapped to a corresponding location in the original image, and its value is estimated using neighboring pixels (Guerreiro, Tomás, Garcia, & Aidos, 2023). Common interpolation methods include Nearest Neighbor, Bilinear, and Bicubic interpolation.

Triwijoyo and Adil (2021) compared these interpolation methods and reported that Bicubic interpolation produced sharper results and lower error rates than Bilinear and Nearest Neighbor methods. Bicubic interpolation considers a 4 × 4 neighborhood of 16 surrounding pixels and assigns higher weights to closer pixels, resulting in improved edge preservation. Although increasing image dimensions raises computational complexity, Bicubic interpolation provides a balanced tradeoff between processing time and output quality.

Several studies standardized input image dimensions, such as 256 × 256 × 3, to reduce computational cost and ensure consistency during model training. The resizing process was commonly performed using Bicubic interpolation due to its improved structural preservation and reduced interpolation error (Triwijoyo & Adil, 2021).

Implications for Paired Image Post-processing Pipeline Design. The reviewed literature indicates that effective paired image post-processing pipelines require structured and coordinated stages to ensure dataset reliability. Image signal processing operations reduce sensor-related artifacts, while geometric and photometric alignment improve spatial and chromatic consistency between sharp and blurred image pairs. Resolution standardization, supported by Bicubic interpolation, preserves structural details while maintaining computational efficiency.

Studies on blur detection further demonstrate that motion blur significantly reduces Laplacian edge responses, supporting the use of Laplacian-based sharpness metrics for objective frame quality assessment. In addition, maintaining temporal and photometric consistency during frame selection ensures that differences between paired images primarily result from motion-induced blur rather than acquisition variability.

Collectively, these findings highlight the importance of a systematic and multi-stage pipeline design to support reliable dataset construction and evaluation in image deblurring research.

Hyperparameter Selection in Deep Learning–Based Motion Deblurring

Hyperparameter selection plays an important role in the performance of deep learning–based motion deblurring models. While recent studies focus on improving network architectures, existing literature shows that training stability and restoration quality are also strongly affected by optimization-related hyperparameters. These include the optimizer, learning rate, normalization strategy, and training configuration. Improper hyperparameter settings can lead to unstable convergence, loss of fine details, and inefficient training, especially when models are adapted to real-world motion blur.

Multi-Scale Motion Deblurring and Training Configuration. Gao et al. (2023) propose an efficient multi-scale network with a learnable discrete wavelet transform, referred to as MLWNet, for blind motion deblurring. Their work addresses limitations of traditional coarse-to-fine schemes by simplifying multi-scale processing through a single-input multiple-output design. The study highlights that, aside from architectural improvements, carefully controlled training hyperparameters are necessary to achieve state-of-the-art (SOTA) performance on real-world and synthetic datasets. The authors employ fixed training settings, including patch size, data augmentation, optimizer choice, and learning rate scheduling, to ensure stable optimization and effective recovery of high-frequency image details.

The training configuration used by Gao et al. (2023) includes a patch size of 256 × 256 pixels and simple data augmentation techniques such as random flips and rotations. These settings increase data diversity while preserving the physical characteristics of motion blur. Their results suggest that consistent training configurations contribute to both computational efficiency and high restoration quality across different datasets.

Optimizer Selection in Deep Neural Networks. Optimizer choice is a critical hyperparameter in deep neural network training. Kingma and Ba (2015) introduce the Adam optimizer, which adapts learning rates for individual parameters using first-order and second-order moment estimates. Adam is shown to be computationally efficient, memory efficient, and suitable for problems with noisy and non-stationary gradients. The authors emphasize that Adam requires minimal manual tuning and performs well in large-scale deep learning tasks.

Building on Adam, Gao et al. (2023) adopt AdamW, a variant that decouples weight decay from gradient updates. This modification improves regularization and helps prevent overfitting during training. The use of AdamW in motion deblurring models supports stable convergence while maintaining high reconstruction accuracy. Recent theoretical and empirical work by Vasudeva, Lee, Sharan, and Soltanolkotabi (2025) further shows that Adam-based optimizers learn richer feature representations than standard gradient descent. Their findings indicate that Adam-trained models generalize better under distribution shifts, which is relevant for motion deblurring tasks involving different real-world blur characteristics.

Learning Rate and Learning Rate Scheduling. The learning rate is a fundamental hyperparameter that controls the magnitude of parameter updates during optimization. According to GeeksforGeeks (2025), a learning rate that is too high can cause divergence or unstable training, while a very low learning rate leads to slow convergence and increased computational cost. An optimal learning rate balances training speed and stability.

In the context of motion deblurring, Gao et al. (2023) initialize training with a learning rate of 1×10^{-3} and apply a cosine annealing schedule to gradually reduce the learning rate during training. This approach allows larger updates in early training stages and finer adjustments in later stages. Such scheduling strategies are commonly used in deep learning to improve convergence behavior and final model performance.

Normalization Strategies in Deep Learning. Normalization techniques are used to stabilize training by controlling the distribution of intermediate activations. Ioffe and Szegedy (2015) introduce batch normalization to address the problem of internal covariate shift in deep neural networks. Their work demonstrates that batch normalization allows higher learning rates, improves gradient flow, and reduces sensitivity to parameter initialization. Batch normalization also acts as a form of regularization, sometimes reducing the need for additional techniques such as dropout.

In multi-scale image restoration models, normalization layers help maintain consistent feature distributions across different scales. Gao et al. (2023) incorporate normalization as part of the model architecture, contributing to stable training and improved restoration quality on real-world datasets.

Hyperparameter Control and Model Generalization. Bengio (2012) provides practical recommendations for gradient-based training of deep architectures and defines hyperparameters as external control variables that must be set before training. The author emphasizes that hyperparameters affecting model capacity and stability should be selected based on out-of-sample data rather than training data alone. The study also highlights the challenges of tuning many hyperparameters and suggests that controlled and validated selection is essential for reliable model performance.

The importance of careful hyperparameter control is reinforced in domain adaptation scenarios, where training data may be limited or differ from the original training distribution. Prior studies indicate that preserving well-established hyperparameter configurations helps prevent degradation of learned representations and unstable convergence during fine-tuning.

Review of Related Studies

This section reviews published research articles, conference papers, dissertation, or thesis related to image deblurring benchmarking and dataset construction.

Deblurring in the Wild: A Real-World Image Deblurring Dataset from Smartphone High-Speed Videos (Mahmud, S. M., Noki, M. M. H., Majumder, P. S., Radi, A. M. A., Sukanto, S. D., Lubaina, A., & Khan, M. M, 2025). The study aimed to develop and introduce a large-scale real-world image deblurring dataset constructed from smartphone slow-motion videos. It also aimed to establish the dataset as a new benchmark for evaluating state-of-the-art image deblurring models under realistic mobile imaging conditions.

To achieve these objectives, the authors introduced the Deblurring in the Wild dataset using an iPhone 15 Pro capable of recording high-speed videos at 240 frames per second with a resolution of 1920 × 1080 pixels. Blurred images were synthesized by averaging 30 temporally aligned frames, corresponding to an effective blur duration of approximately 1/8 second. This duration was selected to preserve realistic motion effects while maintaining photometric consistency and sufficient dynamic range. The temporally centered frame was selected as the sharp ground truth.

The dataset construction assumed approximately uniform motion within the averaging window and did not model complex non-linear motion. This design prioritized reproducibility and scalability over physically accurate blur modeling. The final dataset contained more than 42,000 blurred and sharp image pairs collected from 843 indoor and outdoor scenes.

Because the dataset was captured using a consumer smartphone, it inherently reflected mobile imaging characteristics such as smaller sensor size, computational photography pipelines, built-in stabilization, denoising, tone mapping, and limited dynamic range. These characteristics distinguish it from traditional DSLR-based benchmarks and make it more representative of real-world mobile imaging conditions.

For benchmarking, several state-of-the-art models such as Restormer, MPRNet, NAFNet, FFTFormer, DMPHN, MIMO-UNet++, and AdaRevD were evaluated using PSNR and SSIM. Models were tested without dataset-specific fine-tuning to analyze generalization performance.

The results showed significant performance degradation compared to traditional benchmarks. This indicated a generalization gap when models trained on synthetic or DSLR-based datasets were applied to smartphone images. The study demonstrated that smartphone-based datasets reveal limitations of existing deblurring models in real-world mobile imaging conditions.

Following the observed generalization limitations, the authors recommended the integration of motion-related metadata to support motion-aware deblurring approaches. While the present study does not incorporate motion metadata, it similarly examines motion-related factors by analyzing the influence of hardware stabilization on blur characteristics.

The study is relevant because it highlights generalization gaps in smartphone-based deblurring benchmarks using PSNR and SSIM. However, it relied on a single device and did not isolate hardware stabilization as a variable. The present study addresses this limitation by constructing paired datasets under OIS and non-OIS conditions to analyze the effect of stabilization on motion blur and restoration performance. It further evaluates baseline and domain-adaptive fine-tuning to determine whether stabilization-aware data improves model adaptation.

Assessing the Role of Datasets in the Generalization of Motion Deblurring Methods to Real Images (Carbajal, G., Vitoria, P., Lezama, J., & Musé, P., 2025). The study aimed to propose a simple yet effective method for generating sharp and blurred image pairs that account for non-uniform motion blur, in order to overcome the limitations of existing datasets. Using this procedure, the researchers generated a new dataset called the Segmentation-Based Deblurring Dataset (SBDD). The proposed method enables the generation of virtually unlimited and diverse training pairs that closely mimic realistic blur properties.

To evaluate its effectiveness, existing deblurring architectures were trained using the generated pairs and tested through cross-dataset evaluation on three standard real-world blurred image datasets namely GoPro, RealBlur, and REDS. The dataset was produced using a degradation model combined with a realistic camera-shake trajectory generator. Separate blur kernels were applied to segmented objects and background regions using Mask R-CNN, allowing different motion patterns within a single scene. Soft blending of masks was performed to prevent abrupt boundary artifacts.

The images were first converted to the photon domain before convolution, and illumination augmentation was applied to simulate lighting variations and sensor saturation. The blurred segments were then fused and transformed back to the pixel domain. Several degradation factors were systematically analyzed, including discontinuous kernels to simulate ghosting, variations in kernel size and shape controlled by exposure time and focal length, and the effects of different Camera Response Functions. Additional augmentation strategies were introduced to model saturated pixels and light streak artifacts.

The findings revealed that modern deep deblurring networks tend to overfit benchmark datasets and exhibit limited generalization to real motion-blurred images. The proposed synthetic data generation strategy significantly improved cross-dataset performance. Results further showed that simple convolution-based blur synthesis combined with appropriate augmentation outperformed more complex setups. Uniform blur performed better in scenes with gradual motion changes, while non-uniform blur generation was more effective for dynamic environments.

The study is relevant because it emphasizes the need for realistic training and evaluation data to improve the generalization of motion deblurring networks to real-world images. While Carbajal et al. (2025) addressed dataset limitations through synthetic blur modeling, the present study extends this direction by using directly captured motion-blurred images under controlled conditions and applying domain-adaptive fine-tuning and statistical evaluation to further examine model performance and generalization.

Realistic Blur Synthesis for Learning Image Deblurring (Rim, J., Kim, G., Kim, J., Lee, J., Lee, S., & Cho, S., 2022). The study aimed to address the limitations of existing datasets used for training learning-based image deblurring models. Synthetic datasets were found to lack realism, while real-world datasets had limited scene diversity and were difficult to collect across different camera settings. To overcome these issues, the researchers introduced RSBlur, a dataset composed of real blurred images paired with corresponding sharp image sequences. The study also sought to analyze the differences between real and synthetic blur generation processes. Based on this analysis, the researchers proposed a new blur synthesis pipeline designed to generate more realistic blurred images for training deblurring models.

The researchers developed the RSBlur dataset using a dual-camera system with synchronized shutters. One camera captured a blurred image using long exposure, while the other captured nine consecutive sharp images using short exposure within the same time frame. The fifth sharp image was aligned with the center of the blurred image exposure and was used as the ground-truth reference.

A total of 13,358 blurred images from 697 outdoor scenes were collected and divided into training, validation, and test sets. The captured images were first aligned both geometrically and photometrically to remove inconsistencies between the two camera modules. All images were recorded in camera RAW format and later converted into nonlinear sRGB space through a simplified image signal processing pipeline.

To synthesize realistic blur, the study proposed a multi-stage blur generation pipeline. The process included frame interpolation to smooth motion trajectories, linear-space averaging to simulate light accumulation, saturation synthesis using mask-based estimation, conversion to camera RAW space, noise modeling using Poisson and Gaussian distributions, and application of camera image signal processing. Parameter estimation was performed to replicate real camera noise characteristics.

The researchers evaluated the proposed blur synthesis pipeline by generating synthetic blurred images and using them to train a learning-based deblurring model. The trained models were then tested on real blurred datasets to assess generalization performance. SRN-DeblurNet was used as the primary baseline due to its strong performance and shorter training time. It was trained for 262,000 iterations with data augmentations such as random horizontal and vertical flips and random rotations to improve robustness.

The study revealed that several factors in the blur generation process contribute to the gap between real and synthetic blurred images. These factors include motion discontinuity, nonlinear image processing, pixel saturation, and noise modeling.

The proposed synthesis pipeline produced more realistic blurred images compared to conventional synthetic approaches. Experimental results showed that training deblurring models using images generated by the proposed pipeline improved performance on real-world blurred images. However, the study acknowledged that a performance gap still remained between real datasets and synthesized datasets, indicating the need for further refinement of blur modeling techniques.

The study is relevant to the present study because it supports the need for a controlled and systematic benchmarking framework for real motion deblurring under practical mobile imaging conditions. Similar to the present study, it emphasizes the collection of real blurred–sharp image pairs and the application of explicit geometric, photometric, and color alignment to ensure data consistency. Its use of domain-adaptive fine-tuning, data augmentation, and evaluation through PSNR, SSIM, and statistical significance testing aligns with the present study’s objective of conducting a reliable and experimentally grounded assessment of deblurring model performance.

Real-World Blur Dataset for Learning and Benchmarking Deblurring Algorithms (Rim, J., Lee, H., Won, J., & Cho, S., 2020). The study aimed to construct the first large-scale real-world blur dataset composed of paired blurred and sharp images captured under controlled acquisition settings. It sought to design an image acquisition system capable of capturing geometrically aligned image pairs using a dual-camera setup. In addition, the researchers developed a structured postprocessing pipeline to generate high-quality ground truth images. The study further aimed to benchmark existing deblurring models on the proposed dataset and analyze the effectiveness of training strategies that use real-world data compared to synthetic datasets.

The dataset was constructed using a dual-camera acquisition system composed of two Sony A7RM3 mirrorless cameras equipped with Samyang 14mm F2.8 lenses. A beam splitter setup was employed to allow simultaneous capture of blurred and sharp images of the same scene. A multi-camera trigger ensured synchronized acquisition. The dataset focused on static indoor and outdoor low-light scenes to simulate realistic blur conditions caused by longer exposure times. Both RAW and JPEG image formats were collected. The dataset consisted of 4,556 image pairs per subset across 232 static scenes. Two subsets were created: RealBlur-R for RAW images and RealBlur-J for JPEG images.

For benchmarking purposes, the dataset was divided at the scene level. A total of 182 scenes were randomly assigned for training, while the remaining 50 scenes were reserved for testing. Each training subset contained 3,758 image pairs, including 182 reference pairs, whereas each test subset contained 980 image pairs without reference pairs. This split ensured separation between training and evaluation scenes to maintain fairness in benchmarking.

Image acquisition was performed by simultaneously capturing blurred and sharp images using different shutter speeds. The blurred images were captured using a shutter speed of 1/2 second, while the sharp reference images were captured at 1/80 second under controlled ISO settings. Only static scenes were included to avoid additional motion inconsistencies.

A structured postprocessing pipeline was implemented to ensure high-quality paired data. This pipeline included white balance adjustment, demosaicing, cropping of invalid regions, lens distortion correction, downsampling, and denoising of sharp images using the BM3D algorithm.

Geometric alignment between image pairs was achieved using homography-based alignment followed by phase correlation refinement and centroid-based blur kernel alignment. Photometric alignment was performed through linear intensity transformation to ensure consistency between blurred and sharp images.

For benchmarking, the study evaluated models such as SRN-DeblurNet and DeblurGAN-v2. Performance was measured using PSNR and SSIM. The researchers also compared training strategies using synthetic datasets versus real-world datasets to analyze generalization differences.

The study revealed that models trained solely on synthetic datasets exhibited reduced performance when evaluated on real-world blurred images. In contrast, training with the RealBlur dataset led to significant improvements in PSNR and SSIM on authentic blurred images caused by camera shake and object motion. These findings indicate that realistic blur data plays a critical role in enhancing the generalization capability of learning-based deblurring models. Furthermore, deep learning approaches consistently outperformed traditional kernel-based methods under real-world conditions. Overall, the results emphasize that dataset realism substantially affects restoration quality and the reliability of benchmarking outcomes.

The study acknowledged several limitations and proposed directions for future research. The authors suggested the construction of a dynamic scene dataset for quantitative evaluation of motion deblurring under real motion conditions.They also recommended developing a smartphone-based real-world blur dataset to reflect practical mobile imaging scenarios.

This study is relevant because RealBlur emphasized real-world paired acquisition, precise geometric and photometric alignment, and controlled benchmarking, demonstrating that realistic data improves model generalization. Building on this, the present study will construct a smartphone-acquired dataset under controlled OIS and non-OIS conditions and will evaluate stabilization-aware performance and domain-adaptive fine-tuning in mobile imaging contexts. It will also incorporate controlled motion generation and statistical analysis, extending motion deblurring benchmarking to account for smartphone hardware variability and stabilization effects.

Deep Multi-scale Convolutional Neural Network for Dynamic Scene Deblurring (Nah, S., Kim, T. H., & Lee, K. M., 2017). The study aimed to develop a multi-scale convolutional neural network capable of restoring images degraded by dynamic scene blur. It sought to remove complex spatially varying blur without explicitly estimating blur kernels. In addition, the researchers constructed a large-scale realistic dataset of paired blurry and sharp images using high-speed video acquisition. The study also aimed to achieve improved quantitative and qualitative restoration performance compared to state-of-the-art deblurring approaches available at the time.

The dataset was constructed using a GoPro HERO4 Black camera that captured high-speed videos at 240 frames per second with a resolution of 1280 × 720 pixels. A total of 3,214 blurry-sharp image pairs were generated, with 1,111 pairs reserved for testing, which corresponds to approximately one-third of the total dataset. The remaining image pairs were used for training. Motion blur was synthesized by averaging 7 to 13 consecutive sharp frames extracted from the high-speed videos. Gamma correction with a value of $\gamma = 2.2$ was applied during frame integration, and the middle frame in each sequence was selected as the ground truth sharp image. The study employed a multi-scale convolutional neural network trained in an end-to-end manner. A multi-scale loss function was introduced to improve convergence and restoration quality, and adversarial loss was incorporated to enhance perceptual results. The approach did not rely on explicit blur kernel estimation. Performance evaluation was conducted using PSNR, SSIM, and runtime comparison against prior methods.

The proposed multi-scale CNN achieved higher PSNR and SSIM values compared to previous kernel-based and learning-based deblurring methods. The model demonstrated improved capability in handling spatially varying blur without requiring explicit blur kernel estimation. The multi-scale design contributed to better convergence and enhanced restoration quality. The constructed GoPro dataset enabled realistic modeling of dynamic scene blur while preserving paired supervision. Furthermore, the kernel-free learning approach reduced ringing artifacts commonly observed in traditional kernel-estimation-based techniques.

The study is relevant because it established a foundational benchmark for learning-based motion deblurring using paired real-world data and quantitative evaluation metrics. The work of Nah et al. (2017) demonstrated the effectiveness of training deep models on structured blur–sharp image pairs and provided standardized evaluation measures for performance comparison. However, their dataset was created using a high-speed action camera and synthetic blur generated through frame averaging. In contrast, the present study focuses on a first-hand smartphone-acquired dataset.

A Comparative Study for Single Image Blind Deblurring (Lai, W., Huang, J., Hu, Z., Ahuja, N., & Yang, M., 2016). The study aimed to conduct a comparative evaluation of single image blind deblurring algorithms to address the limitations of existing performance assessments. Specifically, it aimed to construct two large datasets composed of real blurred images and synthetic motion-blurred images. These images were annotated using five attributes: man-made, natural, people or face, saturated, and text.

In addition, the study aimed to perform a large-scale user evaluation involving 13 state-of-the-art single image blind deblurring algorithms. Through statistical analysis of user responses across different image attributes, the study sought to systematically analyze algorithm performance. Furthermore, by comparing results between real and synthetic datasets, the study aimed to generate insights regarding image priors, blur models, evaluation datasets, and quality metrics that were not fully discussed in prior publications.

The researchers constructed a real-world blurred image dataset consisting of 100 images collected from prior studies, online sources, and self-captured photographs. These images were taken using various cameras and settings to reflect real-world conditions. Each image was categorized based on five predefined attributes. To ensure computational feasibility, all images were resized so that the maximum dimension was below 1200 pixels.

For the synthetic dataset, 25 sharp images were collected as ground truth. From these, 100 non-uniform and 100 uniform blurred images were generated. Non-uniform blur was created by recording six-dimensional camera motion trajectories using inertial sensors, followed by applying spatially varying blur kernels and adding Gaussian noise. Uniform blur was synthesized using larger blur kernels generated from simulated camera trajectories. Saturated images were also produced by adjusting intensity ranges before applying blur and clipping values.

To evaluate algorithm performance, the study conducted a large-scale human subject experiment involving 2,100 participants recruited through an online crowdsourcing platform. Each participant completed paired image comparisons, ensuring broad perceptual evaluation across thousands of comparisons. Statistical analysis included agreement testing, Bradley–Terry ranking, convergence analysis to confirm sufficient sample size, and significance testing to group algorithms that were statistically indistinguishable at $\alpha = 0.01$.

The results showed that existing algorithms performed better on synthetic datasets than on real-world blurred images, indicating a gap between controlled experiments and practical applications. The study also revealed that current blur models struggle with saturated regions, non-Gaussian noise, and scenes with large depth variation. In terms of evaluation metrics, information fidelity criterion and visual information fidelity were more aligned with human perception than PSNR and SSIM on synthetic datasets. However, no-reference metrics showed weaker correlation on real images.

This study is relevant because prior research demonstrated that datasets containing only 100 to 200 images are already considered large-scale in single image deblurring evaluation and were sufficient to produce statistically meaningful conclusions. In comparison, the proposed study will utilize 1,000 image pairs consisting of 500 OIS and 500 non-OIS samples, providing a stronger empirical basis and better representation of stabilization conditions. Furthermore, the proposed study requires statistical significance at $\alpha = 0.05$, positive mean improvement, and at least a small effect size ($\eta^2 p \geq 0.01$), with performance evaluated through absolute and percent change and interpreted separately for PSNR and SSIM.

Recording and Playback of Camera Shake: Benchmarking Blind Deconvolution with a Real-World Database (Köhler, Hirsch, Mohler, Schölkopf & Harmeling, 2012). The study aimed to develop a new benchmark dataset for evaluating single image blind deconvolution (BD) algorithms, particularly for non-uniform motion blur caused by real camera shake. It addressed the limitation of existing benchmarks that focus only on uniform blur models. The researchers also aimed to record and analyze real human camera motion using a six-degree-of-freedom setup. Another objective was to validate existing imaging models that simplify camera motion into fewer motion parameters. In addition, the study intended to compare state-of-the-art blind deblurring algorithms under both uniform and non-uniform blur assumptions using the proposed dataset.

The researchers recorded human camera shake using a Vicon motion tracking system with 16 high-speed cameras operating at 500 Hz. A compact camera (Samsung WB600) was used to capture motion trajectories during an exposure time of 1/3 second. Six participants performed natural handheld photography, resulting in 40 recorded trajectories, with 35 valid trajectories after filtering.

To reproduce realistic blur, a Stewart platform (Hexapod) with six degrees of freedom and micrometer-level repeatability was used to replay the recorded trajectories. Blurred images were generated by applying selected trajectories to sharp images in a controlled environment.

For dataset creation, two blur trajectories from each of the six participants were randomly selected and applied to four images. This produced 48 blurry images. The recording setup included an SLR camera equipped with a Canon EF 50mm f/1.4 lens. The camera parameters were ISO 100, aperture f/9.0, and exposure time of 1 second. Images were captured in Canon SRAW2 format (16-bit), converted to 8-bit format, and cropped to 800 × 800 pixel patches.

Ground truth images were also recorded by capturing 167 images along each motion trajectory, plus 30 intermediate images. This allowed precise comparison between restored and reference images.

The performance of blind deconvolution algorithms was measured using PSNR. The similarity between restored images and ground truth images was computed by identifying the maximum PSNR value across the trajectory sequence. The study evaluated both uniform blur models and non-uniform blur models in a comparative analysis.

The findings showed that real camera shake commonly includes rotational motion, which results in non-uniform blur. However, in many cases, the spatial variation of the blur was relatively small.

Some blur kernels appeared approximately uniform, while others clearly exhibited non-uniform characteristics. The dataset enabled objective comparison between algorithms assuming uniform blur and those modeling non-uniform motion.

The study provided a realistic and extensible benchmark dataset with ground truth images, allowing more accurate evaluation of blind deblurring algorithms. The dataset was made publicly available to support further research in motion deblurring.

The study is relevant to the proposed study because it uses hardware-based motion generation to produce physically realistic blur instead of synthetic blur simulation. By recording real human camera shake and replaying it using a six-degree-of-freedom robotic platform, the study generated blur patterns caused by actual translational and rotational motion.

This approach is similar to the proposed study's motion generation strategies. The recorded human shake corresponds to the handshake-based method, where irregular oscillations and small rotations create complex non-uniform blur. The robotic replay of controlled motion is comparable to the vibration-based setup, which induces measurable frequency-based motion. Moreover, translational camera movement aligns with the sliding-based method that produces uniform linear motion blur.

Both studies emphasize controlled physical motion to generate realistic blur patterns, supporting reliable benchmarking of motion deblurring models.

Comparison of Related Studies and the Proposed Study

Table 4 presents a structured analysis of key literature relevant to real-world image deblurring. The table compares each study in terms of research focus, methodology, similarities to the proposed study, and its relation to the proposed work. This comparison highlights existing approaches, identifies methodological gaps, and clarifies how the proposed study extends prior work through controlled smartphone-based acquisition and hardware stabilization analysis.

Table 4. Comparison of Related Studies and the Proposed Study

Title of the Study	Research Focus	Methodology	Similarities to the Proposed Study	Relation to the Proposed Study
Deblurring in the Wild: A Real-World Image Deblurring Dataset from Smartphone High-Speed Videos (Mahmud et al., 2025)	Construction of a large-scale smartphone-based real-world deblurring dataset and benchmarking state-of-the-art models	Frame averaging from 240 fps smartphone videos; 42,000 blurred–sharp pairs; evaluation using PSNR and SSIM; no dataset-specific fine-tuning	Uses smartphone-acquired data; evaluates models using PSNR and SSIM; highlights generalization gaps on mobile data	Relates by emphasizing smartphone realism. However, it uses a single device and does not isolate hardware stabilization. The proposed study isolates OIS vs non-OIS and evaluates domain-adaptive fine-tuning.
Assessing the Role of Datasets in the Generalization of Motion Deblurring Methods to Real Images (Carbajal et al., 2025)	Improving generalization through synthetic non-uniform blur modeling	Segmentation-based blur synthesis using Mask R-CNN; degradation modeling; cross-dataset evaluation	Focuses on generalization gap; uses benchmarking across datasets	Addresses dataset realism through synthetic modeling. The proposed study instead uses first-hand captured real blur under controlled stabilization conditions.
Realistic Blur Synthesis for Learning Image Deblurring (Rim et al., 2022)	Realistic blur synthesis pipeline and RSBlur dataset	Dual-camera capture; RAW alignment; noise modeling; training SRN-DeblurNet; PSNR and SSIM evaluation	Uses real blurred–sharp pairs; applies geometric and photometric alignment; supports domain adaptation	Supports controlled acquisition and alignment principles adopted in the proposed study
Real-World Blur Dataset for Learning and Benchmarking Deblurring Algorithms (Rim et al., 2020)	First large-scale real-world blur dataset (RealBlur) using dual-camera DSLR setup	Beam splitter dual-camera system; geometric and photometric alignment; benchmarking via PSNR and SSIM	Emphasizes real paired acquisition and controlled benchmarking; applies geometric and photometric alignment; scene-level dataset split	Establishes importance of real data for generalization. The proposed study extends this to smartphone-based capture under OIS and non-OIS conditions.

Table 4. Continued…

Title of the Study	Research Focus	Methodology	Similarities to the Proposed Study	Relation to the Proposed Study
Deep Multi-scale CNN for Dynamic Scene Deblurring (Nah et al., 2017)	Multi-scale CNN for dynamic scene blur removal and GoPro Dataset	High-speed GoPro capture; frame averaging; end-to-end training; PSNR and SSIM evaluation	Uses paired real-world dataset; quantitative evaluation metrics consistent with the proposed study	Provides a foundational benchmark dataset. However, blur was generated through frame averaging and does not represent real single-exposure blur
A Comparative Study for Single Image Blind Deblurring (Lai et al., 2016)	Comparative evaluation of blind deblurring algorithms	Construction of real and synthetic blur datasets; human subject evaluation; statistical ranking and significance testing	Uses structured comparative evaluation; highlights performance gap between synthetic and real blur datasets; applies statistical analysis to interpret results	Supports the need for statistically grounded benchmarking. The proposed study extends this by isolating hardware stabilization and model adaptation effects using a controlled factorial design.
Recording and Playback of Camera Shake (Köhler et al., 2012)	Hardware-based recording and replay of real camera shake	Motion tracking system; robotic platform replay; PSNR-based evaluation	Uses physically generated blur rather than synthetic kernels	Conceptually similar in controlled motion generation. The proposed study adapts physical motion generation (handshake, vibration, sliding) within smartphone-based capture.

Comparison of Benchmark Datasets and the Proposed Dataset

Table 5 presents a comparative analysis of existing motion deblurring benchmark datasets and the proposed stabilization-aware dataset. The comparison highlights differences in device setup, blur generation methods, alignment procedures, dataset scale, and evaluation strategies.

Table 5. Comparison of Motion Deblurring Benchmark Datasets and the Proposed Dataset

Dataset	Device & Capture Setup	Blur Generation	Alignment & Preprocessing	Size & Evaluation
Deblurring in the Wild (Mahmud et al., 2025)	iPhone 15 Pro; 240 fps; 1920 × 1080; smartphone-based capture	Blur synthesized by averaging 30 aligned frames (~1/8 sec exposure); centered frame used as sharp reference	Assumed near-uniform motion; limited nonlinear motion modeling; retained smartphone ISP characteristics	>42,000 pairs from 843 scenes; PSNR and SSIM; cross-dataset generalization analysis
SBDD (Carbajal et al., 2025)	Synthetic generation using segmentation-based modeling; Mask R-CNN object separation	Separate blur kernels applied to segmented regions; camera-shake trajectory generator; non-uniform synthetic blur	Photon-domain convolution; soft mask blending; illumination augmentation; CRF and saturation modeling	Virtually unlimited synthetic pairs; cross-dataset evaluation (GoPro, RealBlur, REDS); PSNR and SSIM
RSBlur (Rim et al., 2022)	Dual-camera synchronized shutters; long exposure blurred image and short exposure sharp sequence	Real blurred capture; 9 sharp frames captured; fifth frame as ground truth; realistic synthetic augmentation pipeline	Geometric and photometric alignment; RAW-to-sRGB processing; Poisson and Gaussian noise modeling; saturation modeling	13,358 blurred images; evaluated using SRN-DeblurNet; PSNR and SSIM

Table 5. Continued…

Dataset	Device & Capture Setup	Blur Generation	Alignment & Preprocessing	Size & Evaluation
RealBlur-R & RealBlur-J (Rim et al., 2020)	Dual Sony A7RM3 cameras with beam splitter; synchronized capture	Real blur via shutter speed difference (1/2s blurred, 1/80s sharp); static scenes	Homography alignment; phase correlation refinement; centroid-based kernel alignment; photometric linear transformation	RealBlur-R: 4,556 image pairs; RAW pipeline benchmarking. RealBlur-J: 4,556 image pairs; ISP-processed JPEG benchmarking. Evaluation using PSNR and SSIM; scene-level train/test split
GoPro Dataset (Nah et al., 2017)	GoPro HERO4 Black; 240 fps; 1280 × 720	Synthetic blur via averaging 7–13 frames; gamma correction ($\gamma = 2.2$); middle frame as ground truth	Temporal alignment through frame averaging; no complex geometric correction reported	3,214 blur–sharp pairs; PSNR and SSIM; multi-scale CNN benchmarking
Lai Dataset (Lai et al., 2016)	Mixed real-world images from various cameras; synthetic blur using simulated trajectories	Uniform and non-uniform synthetic blur; Gaussian noise; saturation modeling	Resized images (<1200 px); user-study-based perceptual evaluation; attribute-based categorization	100 real blurred images; 200 synthetic images; large-scale user evaluation; perceptual ranking analysis
Köhler Dataset (2012)	Samsung WB600 with Vicon motion tracking (500 Hz); Stewart platform (Hexapod) for replay; SLR with Canon EF 50mm f/1.4	Real human camera shake recorded and physically replayed; non-uniform blur from translational and rotational motion	Ground truth captured along motion trajectory; cropped to 800 × 800 patches; PSNR computed across trajectory alignment	48 blurred images; PSNR-based blind deconvolution benchmarking

Table 5. Continued…

Dataset	Device & Capture Setup	Blur Generation	Alignment & Preprocessing	Size & Evaluation
Proposed StaBlur & UnstaBlur Dataset (Present Study)	Two smartphones from same manufacturer; one with OIS (Xiaomi Poco X6 5G), one without OIS (Xiaomi Redmi Note 13); dual-mounted rigid platform; RAW video capture; 1080 × 1080; 30 fps	Real motion blur generated via sliding (uniform), handshake (non-uniform), and vibration (37.7–65.2 Hz); blur extracted directly from RAW frames	Laplacian-based sharpness selection; geometric alignment; photometric alignment; color normalization; bicubic downsampling to 256 × 256; structured dataset partitioning	1,000 paired images (500 OIS, 500 non-OIS); balanced motion categories; PSNR and SSIM; Two-Way ANOVA; effect size ($\eta^2 p$); domain-adaptive fine-tuning evaluation

Synthesis

The reviewed literature establishes that motion blur is a persistent and complex form of image degradation that significantly affects the performance of computer vision systems across multiple real-world applications. Motion blur arises from camera movement, object motion, exposure settings, and hardware characteristics during image capture. Traditional image restoration methods rely on simplified mathematical models and explicit blur kernel estimation, which often fail under real-world conditions due to non-uniform and spatially varying blur patterns. These limitations highlight the need for more adaptive and data-driven approaches.

Recent advances in deep learning have led to substantial improvements in motion deblurring performance. Learning-based methods, particularly supervised and blind deblurring models, have demonstrated strong capability in recovering fine image details without explicit blur kernel modeling. Among different architectures, Transformer-based models consistently outperform CNN-, RNN-, and GAN-based approaches in both pixel-level accuracy and structural similarity. Their ability to model long-range dependencies and global contextual information makes them especially suitable for handling complex motion blur.

Despite these advances, existing benchmarking practices remain limited. Many deblurring models are trained and evaluated using synthetic blur datasets, which often fail to capture the physical image formation process of real-world motion blur. As a result, models trained on synthetic data frequently show degraded performance when applied to real images. Although real-world datasets improve realism, they are commonly constrained by limited blur diversity or specialized acquisition systems that do not reflect consumer imaging conditions.

The literature further reveals that hardware-based stabilization, particularly OIS, alters motion blur characteristics during image capture. While OIS improves visual quality, it changes the underlying motion patterns recorded by the sensor. Existing datasets and studies primarily focus on stabilization or motion compensation rather than explicitly analyzing how OIS influences motion deblurring model performance. There is limited availability of datasets that systematically compare images captured with OIS and without OIS under comparable conditions.

In addition, dataset construction and post-processing are shown to be critical factors in reliable benchmarking. Prior studies emphasize the importance of controlled image acquisition, geometric and photometric alignment, noise handling, and objective sharpness assessment. Laplacian-based measures are widely supported as effective indicators for distinguishing sharp and blurred frames, ensuring consistent paired data selection.

Overall, the literature indicates a clear research gap in stabilization-aware benchmarking for motion deblurring. There is insufficient empirical analysis of how deep learning–based deblurring models, particularly Transformer-based architectures, generalize across stabilized and unstabilized capture conditions. This gap justifies the need for a dedicated dataset construction strategy that includes both OIS and non-OIS images captured using consumer mobile devices. Addressing this gap will enable more reliable evaluation of model robustness and provide clearer insights into real-world motion deblurring performance, forming the basis for the proposed study.

Conceptual Framework

This study will examine the effect of hardware-based stabilization and domain-adaptive fine-tuning on motion deblurring performance using a transformer-based model. The conceptual framework follows the Input–Process–Output (IPO) model as shown in Figure 1.

Input. The input of the study will consist of image acquisition resources, a pre-trained transformer-based model, and evaluation metrics. Image data will be captured using smartphone devices with OIS and without OIS. Controlled motion induction setups, such as hand-induced shake, vibration-based motion, and sliding translational movement, will be used to generate realistic motion blur conditions. The pre-trained MLWNet will serve as the baseline model. Performance will be measured using PSNR and SSIM.

Process. The process will be divided into five major phases. First, dataset construction will be conducted through RAW image collection, frame extraction, alignment, bicubic interpolation, and organization into OIS and Non-OIS subsets. The dataset will then be split into training, validation, and testing sets. Second, baseline benchmarking will be performed by evaluating the pre-trained MLWNet on the testing subsets using PSNR and SSIM. Third, domain-adaptive fine-tuning will be carried out by retraining the model using the constructed dataset. Fourth, post-fine-tuning benchmarking will be conducted using the same evaluation metrics. Finally, statistical analysis will be applied using Two-Way ANOVA to analyze the effects of stabilization condition, model configuration, and their interaction.

Output. The output of the study will include a curated real-world paired dataset composed of StaBlur (OIS subset) and UnstaBlur (Non-OIS subset). Quantitative restoration results will be reported using baseline and fine-tuned PSNR and SSIM scores. Analytical findings will describe performance differences under different stabilization conditions and model configurations. Statistical evidence will determine whether hardware stabilization and domain-adaptive fine-tuning produce significant improvements in motion deblurring performance.

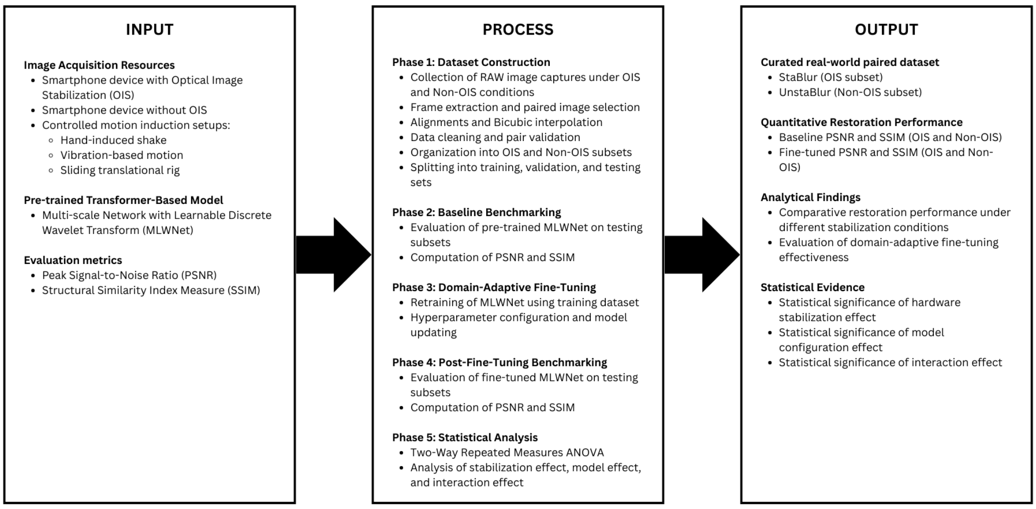

Figure 1. Conceptual Framework

Figure 1 illustrates the relationship between the input variables, methodological procedures, and expected outputs of the study. It presents how hardware stabilization conditions and model adaptation strategies will influence quantitative restoration performance and statistical findings.

METHODOLOGY

This chapter presents the methodological framework developed to achieve the objectives of the study and address the research gaps identified in Chapter 1. The study adopts an experimental approach to examine how OIS influences the performance of a transformer-based motion deblurring model using real-world image data. The methodology is designed to ensure systematic data collection, objective performance evaluation, and statistically grounded analysis.

Research Design

This study will employ a quantitative experimental research design focused on the systematic benchmarking of a deep learning–based motion deblurring model under controlled image capture conditions. The quantitative approach will be used because the study relies on numerical image quality metrics and statistical analysis to evaluate model performance.

The experiment will evaluate the restoration performance of a single transformer-based model, the MLWNet, using real-world image data captured with and without OIS. Performance evaluation will be based on objective quantitative measures, namely PSNR and SSIM.

Overview of MLWNet Architecture

The motion deblurring model used in this study is MLWNet, a transformer-based multi-scale network designed for blind motion deblurring. MLWNet adopts a single-input multiple-output architecture that restores images progressively at multiple scales, reducing the complexity of traditional coarse-to-fine designs while maintaining consistency across resolutions.

As illustrated in Figure 2, the model combines encoder blocks with wavelet-based feature decomposition and fusion mechanisms to capture both global structure and fine-grained details. The integration of learnable wavelet operations enables effective separation and reconstruction of low- and high-frequency information, which is important for handling real-world motion blur. In this study, the original MLWNet architecture is used without modification, and all experiments focus on benchmarking and domain-adaptive fine-tuning rather than architectural changes.

The methodology will involve constructing a curated dataset of paired motion-blurred and sharp images captured from the same scenes using two different smartphone devices under controlled capture conditions. The dataset will be used to evaluate the baseline performance of a pre-trained MLWNet model and to assess performance changes after domain-adaptive fine-tuning. A two-way ANOVA will be used to evaluate the main and interaction effects of hardware stabilization condition (OIS and non-OIS) and model configuration (pre-trained and fine-tuned) on motion deblurring performance.

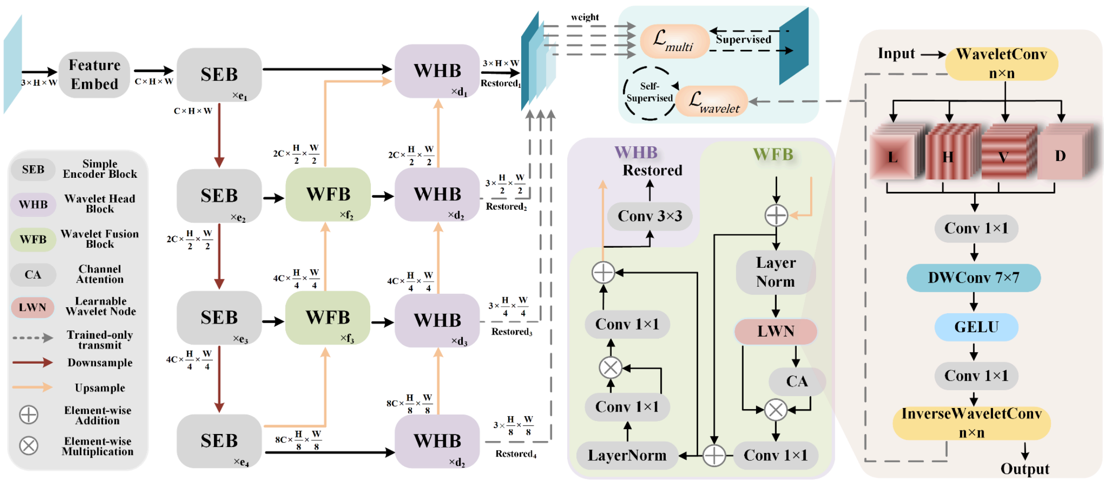

Figure 2. MLWNet Architecture Overview (Gao et al., 2023)

Methodology Framework

Figure 3 illustrates the proposed experimental process flow of the study. The methodology is organized into five sequential phases: (1) dataset construction, (2) baseline benchmarking, (3) model fine-tuning, (4) post-fine-tuning benchmarking, and (5) statistical analysis.

The process begins with the construction of a curated dataset composed of synchronized RAW video captures under OIS and non-OIS conditions. The dataset will undergo cleaning, validation, and partitioning into training, validation, and testing sets.

Next, baseline benchmarking will be conducted using a pre-trained MLWNet model to obtain initial performance scores in terms of PSNR and SSIM.

The model will then undergo supervised domain adaptive fine-tuning using the training and validation sets. After fine-tuning, benchmarking will be repeated under identical conditions to ensure fair comparison.

Finally, statistical analysis using ANOVA will be performed to evaluate the significance of performance differences across OIS and non-OIS conditions, as well as before and after fine-tuning.

Each phase builds upon the outputs of the previous phase to ensure a systematic, controlled, and replicable evaluation of motion deblurring performance.

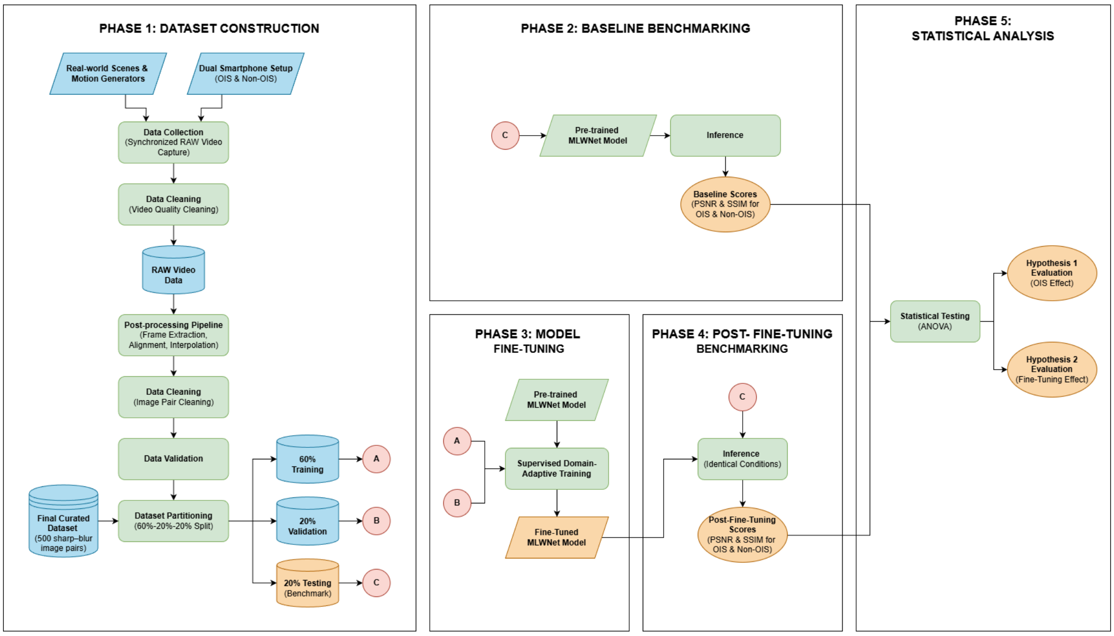

Figure 3. Methodology Framework

Phase 1: Dataset Construction

Phase 1 will focus on the construction of a curated real-world dataset consisting of paired motion-blurred and sharp images captured under different hardware stabilization conditions. This phase is critical because the quality and consistency of the dataset will directly affect the validity of subsequent benchmarking and statistical analysis.

Figure 4 illustrates the proposed dataset construction process adopted in this study. The process is adapted from the dataset construction framework presented by Orr and Crawford (2024) and modified to suit the requirements of real-world motion deblurring under OIS and non-OIS conditions. The adapted process emphasizes first-hand data acquisition, data cleaning, validation, and structured dataset preparation to support reliable benchmarking and statistical analysis.

The dataset construction process will follow a structured sequence of steps, as illustrated in the proposed process flow, to ensure controlled data acquisition, consistency, and reliability.

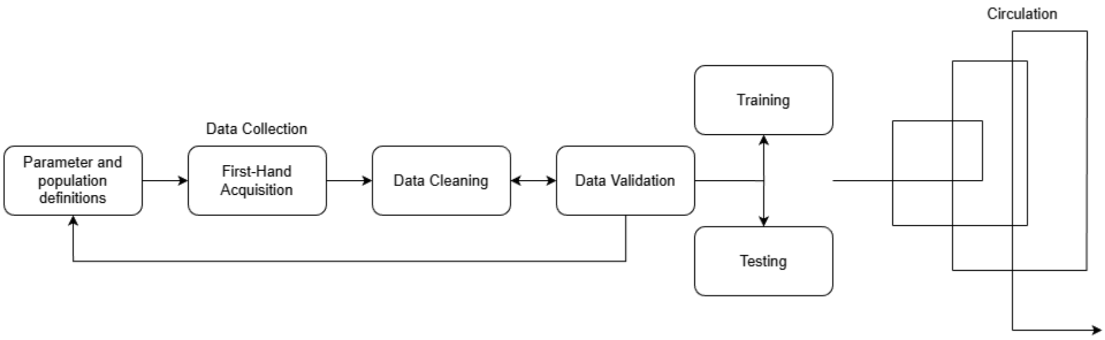

Figure 4. Dataset Construction (adapted from Orr and Crawford, 2024)

Parameter and Population Definition

Prior to first-hand data collection, all image capture parameters and dataset population characteristics will be explicitly defined to ensure consistency, reproducibility, and controlled comparison across stabilization conditions. The dataset will be constructed entirely from first-hand recordings captured using smartphone cameras under real-world imaging conditions.

The primary data source will consist of RAW video recordings, from which individual image frames will be extracted to form motion-blurred inputs and corresponding sharp reference images. All capture parameters will be fixed or locked during recording to prevent automatic adjustments that may introduce unintended variability across frames and devices. These predefined parameters will help ensure that observed performance differences in later experimental phases can be attributed primarily to stabilization conditions and model behavior.

Table 6 presents the image capture parameters used for dataset construction in this study. The parameters define the recording configuration, stabilization settings, and environmental constraints applied during video acquisition.

Table 6. Image Capture Parameters

Parameter	Setting / Description
Number of Devices	Two smartphones from the same manufacturer
Stabilization Capability	One smartphone with OIS; one smartphone without OIS
Primary Lens Selection	Primary wide-angle lens (25mm equivalent focal length) only
Capture Mode	Video recording
File Format	RAW video format
Video Duration	Maximum of five seconds per recording
Resolution	1080 × 1080 pixels
Frame Rate	30 frames per second (fps)
Exposure Mode	Locked; auto-exposure disabled
Shutter Speed	Manually fixed during recording
ISO	Manually fixed during recording
Focus Mode	Locked before recording
White Balance	Locked before recording
Electronic Image Stabilization (EIS)	Disabled
Digital Video Stabilization	Disabled

The dataset population will consist of real-world images of inanimate scenes and objects captured under practical mobile photography conditions. Selected scenes will represent common indoor and outdoor environments and will include a range of textures and structural details to reflect varied visual characteristics encountered in everyday imaging scenarios.

The study population is limited to images captured using two different smartphone models from the same manufacturer, with one device equipped with OIS and the other without OIS. This design introduces controlled hardware variation while isolating the stabilization mechanism as the primary experimental variable. The study does not aim to represent the full diversity of smartphone imaging systems.

Human subjects, facial imagery, and personally identifiable information will be excluded from the dataset. Environmental factors such as background composition and lighting direction will not be fully controlled to preserve real-world variability. However, extreme conditions, particularly low-light or dim environments, will be excluded to avoid introducing confounding degradations unrelated to motion blur.

Through this process, a total of 1,000 image pairs will be constructed and organized into two stabilization-based datasets: StaBlur, which will contain images captured using OIS, and UnstaBlur, which will contain images captured without OIS. Each dataset will consist of 500 image pairs. Within both StaBlur and UnstaBlur, the dataset will include 168 handshake-based image pairs, 166 vibration-based image pairs, and 166 sliding-based image pairs. The detailed distribution of image pairs per stabilization condition is presented in Table 7. Distribution of Image Pairs per Stabilization Condition. This structured composition will ensure balanced representation of motion generation categories across stabilization conditions and will support controlled comparison and paired statistical analysis of model performance.

Table 7. Distribution of Image Pairs per Stabilization Condition

Motion Blur Category	**StaBlur (OIS)**	**UnstaBlur (Non-OIS)**	**Total per Category**
Handshake-based	168	168	336
Vibration-based	166	166	332
Sliding-based	166	166	332
Total per Dataset	**500**	**500**	**1,000**

Data Collection

First-hand image acquisition. First-hand image acquisition will be conducted through a controlled yet practical data collection process designed to capture real-world motion blur and corresponding sharp reference images under different hardware stabilization conditions. The dataset will be constructed entirely from recordings captured by the researchers using smartphone cameras under realistic mobile photography settings.

To ensure consistency, repeatability, and synchronized capture across devices, a series of custom-built acquisition setups will be constructed using readily available materials. These setups will enable controlled camera motion, synchronized triggering, and stable camera positioning while preserving the characteristics of naturally occurring motion blur.

Dual-Smartphone Capture Configuration. A dual-smartphone mounting configuration will be used as the base capture setup, as illustrated in Figure 5. Two smartphones, one equipped with OIS and one without OIS, will be mounted simultaneously on a shared rigid platform. The smartphones will be positioned such that their camera lenses are aligned as closely as possible along the same viewing direction. Both devices will be supported by a single solid base to ensure that any induced motion affects both smartphones consistently. This configuration will allow paired data collection under OIS and non-OIS conditions from the same scene and motion pattern.

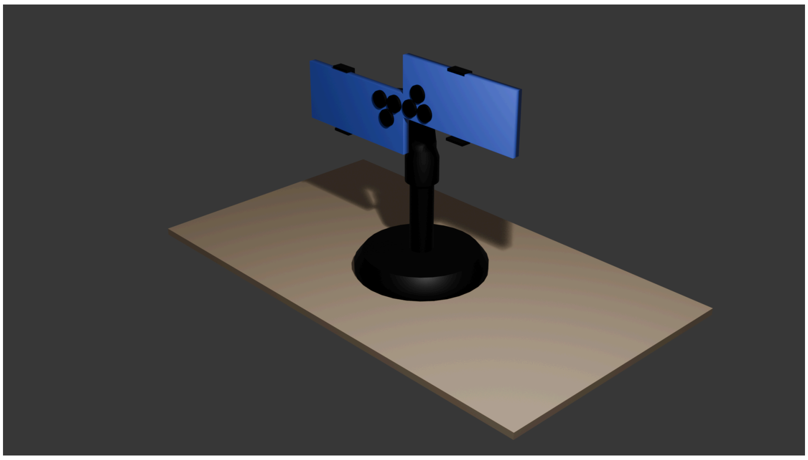

Figure 5. Dual-Smartphone Setup

Motion Generation Setups. To capture diverse and realistic motion blur patterns, multiple motion generation strategies will be employed using modified versions of the base dual-smartphone setup. These strategies are designed to simulate common sources of camera motion encountered during handheld mobile photography.

A sliding-board setup will be constructed by mounting the dual-smartphone platform onto a guided rail system, as shown in Figure 6. Wheels installed beneath the platform and a guiding frame will restrict movement to a single horizontal direction. This setup will allow smooth and repeatable linear motion during recording.

The sliding motion will be manually induced by gently pushing the platform along the guided direction while video recording is active. This strategy will generate uniform translational motion blur suitable for controlled benchmarking.

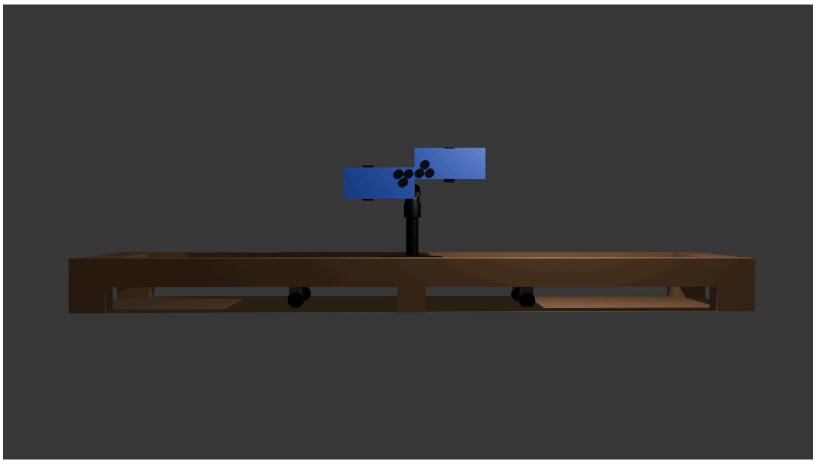

Figure 6. Sliding-Based Motion Generation

To simulate high-frequency camera shake, a vibration-based setup will be constructed, as illustrated in Figure 7. A percussive massage gun will be rigidly mounted using a clamp and pressed against the base platform to induce controlled linear vibration. Rubber isolation feet will be installed beneath the platform to reduce vibration transfer to external surfaces and improve experimental stability.

Motion intensity will not be defined using manufacturer-defined speed labels. Instead, vibration frequency will be measured using an accelerometer sensor application, and vibration intensity will be based on measured frequency values in Hertz (Hz).

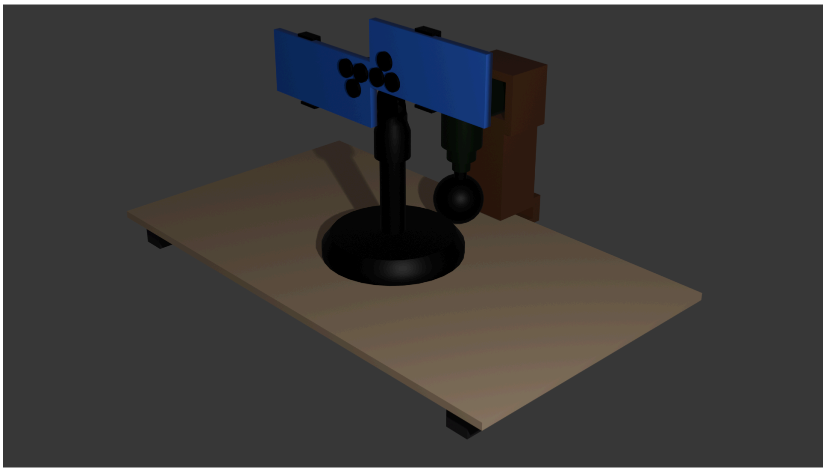

Figure 7. Vibration-Based Motion Generation

Table 8 presents the measured frequency range of the device.

Table 8. Measured Low-Frequency Vibration Range

Classification	Frequency Range (Hz)
Low-Frequency Vibration	37.7 – 65.2

These predefined frequency ranges will enable systematic evaluation of model performance across increasing motion severity while maintaining controlled and repeatable capture conditions.

Motion Generation Strategies and Resulting Blur Types. Different motion generation strategies will be used to capture a wide range of real-world blur patterns. Table 9 summarizes the strategies, their implementation, and the expected resulting blur characteristics.

Table 9. Motion Generation Strategies

Blur Type	Strategy	Implementation	Resulting Blur Pattern
Uniform Motion	Sliding-board setup	A dual-smartphone board will be gently pushed in one straight direction along a guided rail during recording.	Consistent linear streaks (linear motion blur).
Non-uniform Motion (I)	Hand-shake method	Researchers will manually shake the mounted camera setup during recording to introduce irregular motion.	Complex blur patterns resembling natural camera shake.
Non-uniform Motion (II)	Vibration-based setup	A mounted vibration device will induce controlled low-frequency motion during image capture.	Dense, fine-grained blur patterns with increasing severity.

Materials Used for Data Collection. Table 10 lists the materials that will be used for each capture setup constructed in this study.

Table 10. Materials to be Used for Data Collection Setups

Setup	Materials
Dual smartphone setup	Two smartphones (with and without OIS), phone holder, wooden board (platform)
Sliding setup	Two smartphones, phone holder, wooden board, four wheels, 2×2 wood frame, plywood base
Vibration-based setup	Two smartphones, phone holder, wooden board, massage gun, clamp, four rubber feet

Paired Image Post-processing Pipeline

To ensure reliable construction of paired sharp and motion-blurred images, a structured paired image post-processing pipeline will be applied to all recorded video data. The pipeline is designed to minimize inconsistencies arising from device-specific imaging characteristics, camera misalignment, and photometric variation, while preserving motion blur as the primary source of degradation. Without such processing, performance evaluation of motion deblurring models may be influenced by dataset artifacts rather than true motion-induced blur.

Figure 8 illustrates the proposed paired image post-processing pipeline adopted in this study. The pipeline consists of sequential stages, beginning with frame extraction and candidate selection, followed by alignment and normalization procedures, and concluding with dataset structuring.

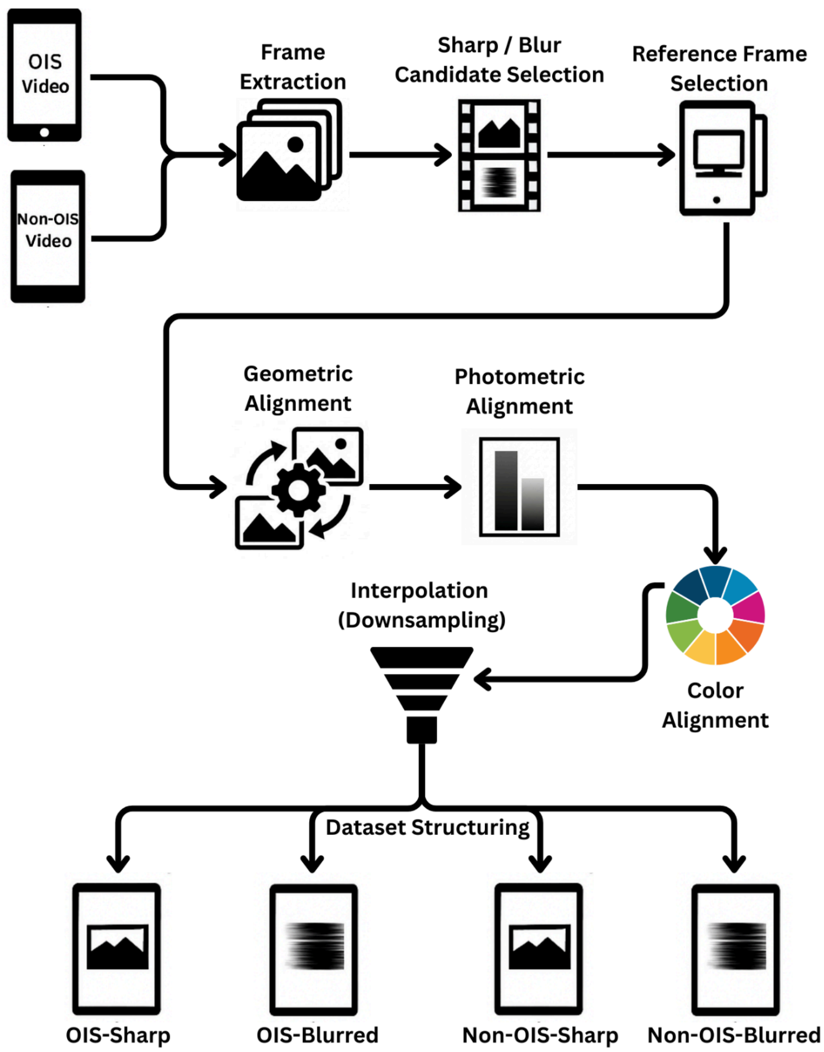

Figure 8. Paired Image Post-Processing Pipeline

Frame Extraction from OIS and Non-OIS Videos. The pipeline will begin with frame extraction from RAW video recordings captured using OIS-enabled and non-OIS smartphones. Individual frames will be extracted at the native recording frame rate to preserve temporal continuity and motion characteristics.

Video-based acquisition will allow both sharp and motion-blurred frames to be selected from the same recording sequence, ensuring that paired images originate from identical scenes and capture conditions. This approach will reduce scene-level variability and improve pairing consistency.

Sharp and Blur Candidate Selection. From the extracted frames, candidate sharp and blurred frames will be identified. Candidate selection will be guided by objective sharpness assessment rather than subjective visual inspection.

Laplacian-based sharpness measures will be applied to each frame to quantify high-frequency content. Frames exhibiting higher Laplacian responses will be considered sharp candidates, while frames with significantly reduced responses will be treated as motion-blurred candidates. This criterion is based on the observation that motion blur suppresses edge strength and high-frequency details.

Using objective sharpness measures will ensure consistent and repeatable identification of sharp reference frames and motion-blurred inputs across different scenes, devices, and motion conditions.

Reference Frame Selection. For each scene and stabilization condition, a single sharp reference frame will be selected from the pool of sharp candidates. This reference frame will serve as the ground truth image for evaluating motion deblurring performance.

Selecting a stable reference frame prior to alignment and normalization is critical, as all subsequent correction steps will be performed relative to this frame. This step ensures that paired images differ primarily due to motion blur rather than temporal or photometric inconsistencies.

Geometric Alignment. Following reference frame selection, geometric alignment will be applied to correct spatial misalignment between paired images. Minor misalignments may arise due to differences in camera lens positioning, mechanical tolerances in the capture setup, and timing differences during recording.

Geometric alignment will aim to achieve pixel-level correspondence between sharp and blurred frames by compensating for translation, rotation, and small perspective deviations. Correcting geometric misalignment is essential to prevent structural discrepancies from being interpreted as restoration errors during model evaluation.

Photometric Alignment. After geometric alignment, photometric alignment will be performed to reduce intensity differences between paired images. Even under fixed capture parameters, variations in exposure response and sensor behavior may lead to brightness and contrast mismatches between frames.

Photometric alignment will adjust intensity statistics to match the reference frame while preserving spatial structure. Performing photometric correction after geometric alignment ensures that intensity adjustments are applied to spatially corresponding regions, improving consistency across pairs.

Color Alignment. Color alignment will then be applied to correct chromatic inconsistencies between paired images. Differences in sensor response, lens characteristics, and internal image processing pipelines may cause color shifts even between recordings captured under similar conditions.

Color alignment will normalize color distributions between paired images to ensure consistent chromatic appearance. This step is critical for preventing color discrepancies from influencing restoration metrics and model behavior, particularly for learning-based deblurring approaches sensitive to color variation.

Interpolation (Downsampling). After alignment procedures, interpolation will be applied to standardize image resolution across all paired samples. Resolution adjustment will reduce high-frequency noise, suppress minor parallax inconsistencies, and decrease computational complexity during model training.

Images will be resized to a fixed dimension of 256 × 256 using Bicubic interpolation. Bicubic interpolation considers a 4 × 4 neighborhood of surrounding pixels and preserves edge structures more effectively than Nearest Neighbor and Bilinear methods.

Standardizing image dimensions will ensure structural consistency between sharp and blurred pairs while maintaining fair and efficient evaluation of motion deblurring models.

Operational Preprocessing and Dataset Organization

After alignment and interpolation, additional preprocessing and organizational steps will be applied to prepare the dataset for benchmarking and model fine-tuning. These steps focus on data consistency, storage reliability, and experimental clarity rather than artifact correction.

Table 11 summarizes the preprocessing and organization steps that will be applied during dataset preparation.

Table 11. Image Preprocessing and Organization Steps to be Applied During Dataset Preparation

Step	Action / Transformation	Goal	Implementation Detail
Pixel normalization	Normalize pixel values to a fixed range	Ensure consistent numerical scale for model processing	Pixel values scaled to the range [0, 1]
Format standardization	Convert extracted frames to a lossless format	Prevent compression artifacts during storage and reuse	All frames stored in PNG format

Table 11. Continued…

Step	Action / Transformation	Goal	Implementation Detail
Data augmentation	Apply light augmentation (e.g., horizontal flipping)	Improve generalization during model fine-tuning	Applied only to the fine-tuning dataset, not baseline evaluation data
Directory structuring	Organize files into a standardized folder hierarchy	Enable reliable pairing and dataset management	A consistent folder structure will be used to group images by scene and capture condition
Naming convention	Apply a consistent file naming scheme	Ensure correct sharp–blur pair association	File names will encode scene identifiers and stabilization conditions

Data Cleaning

Data cleaning will be conducted to ensure that only high-quality, reliable video recordings and image pairs are included in the dataset. This step is necessary to remove acquisition artifacts, recording errors, and inconsistent samples that may negatively affect frame pairing, post-processing alignment, and subsequent benchmarking results. Without systematic data cleaning, dataset noise and unintended artifacts may introduce bias and reduce the validity of quantitative evaluation.

The data cleaning process will be performed in two stages: video-level quality cleaning and image-pair-level cleaning. Each stage will follow predefined inclusion and exclusion criteria to ensure consistency and reproducibility.

Video Quality Cleaning. At the video level, all recorded RAW video clips will be inspected to verify recording integrity and suitability for frame extraction. Only videos that meet all quality requirements will be retained for further processing.

Table 12 summarizes the checklist used for video quality cleaning.

Table 12. Video Quality Cleaning Checklist

#	Checklist Item	Description / Criterion
1	File completeness	Video file is fully recorded and not truncated.
2	Decode validity	RAW video can be decoded without frame loss or errors.
3	Continuous recording	No unintended pauses or recording interruptions.
4	Scene relevance	Video content matches the target scene and capture task.
5	Exposure stability	No extreme flicker or sensor saturation across frames.
6	Focus reliability	Video is not globally out of focus throughout the clip.
7	Camera handling artifacts	No severe accidental shake unrelated to intended motion generation.
8	Duration sufficiency	Video length is sufficient for required frame pairing.
9	Inclusion decision	Video is accepted or rejected for frame extraction.

Videos failing any of the above criteria will be excluded entirely to prevent propagation of recording-level errors into the frame extraction and pairing stages. This conservative exclusion strategy ensures that only stable and representative recordings are used during dataset construction.

Image Pair Cleaning. After frame extraction and candidate selection, image-pair-level cleaning will be performed to ensure that each sharp–blurred pair is valid, consistent, and suitable for benchmarking and model training. This step focuses on enforcing temporal, spatial, and photometric consistency between paired images.

Objective sharpness assessment will be used as the primary criterion for distinguishing sharp reference frames from motion-blurred frames. Specifically, the variance of the Laplacian will be computed for candidate frames, as reduced

Laplacian responses indicate motion blur due to suppression of high-frequency image content.

Table 13 presents the checklist used for image pair cleaning.

Table 13. Image Pair Cleaning Checklist

#	Checklist Item	Description / Criterion
1	Candidate frame pooling	Frames are sampled from the same RAW video segment.
2	Temporal proximity rule	Frames used in a pair are temporally adjacent or within a fixed window.
3	Sharpness scoring	Variance of Laplacian is computed for each candidate frame.
4	Sharp frame selection	Frames with high Laplacian variance are selected as sharp references.
5	Blurred frame selection	Frames with low Laplacian variance are selected as motion-blurred counterparts.
6	Threshold consistency	Fixed or statistically derived thresholds are applied consistently.
7	Pair formation integrity	Selected sharp and blurred frames originate from the same scene instance.
8	Resolution and color consistency	Paired images share identical resolution and color space.
9	Redundancy control	Near-duplicate sharp–blur pairs are removed.
10	Pair locking	Finalized pairs are fixed before validation and training.

Once an image pair satisfies all checklist criteria, it will be locked and excluded from further modification. This ensures that downstream processing, validation, and benchmarking operate on a fixed and well-defined dataset.

Data Validation

After data cleaning and paired image post-processing, a multi-stage data validation process will be conducted to ensure that each finalized sharp–blurred image pair satisfies strict consistency and correspondence requirements. Data validation is necessary to confirm that paired images originate from the same scene instance, exhibit consistent motion characteristics, and differ primarily due to motion-induced blur rather than acquisition artifacts or scene changes.

The validation process will follow a multi-stage validation framework, combining objective computational checks with limited manual inspection. Each validation stage targets a specific source of potential inconsistency and is applied sequentially to all candidate image pairs.

Multi-Stage Validation Procedure. The validation framework consists of multiple stages addressing sharpness, correctness, temporal consistency, spatial correspondence, motion coherence, and photometric and color consistency. Image pairs that fail any validation stage will be excluded from the dataset.

Table 14 summarizes the validation stages, methods used, and their corresponding validation objective.

Table 14. Multi-Stage Data Validation Framework

#	Validation Stage	Method Used	What It Validates
1	Sharpness validation	Variance of Laplacian	Distinguishes sharp and blurred frames.
2	Temporal consistency	Laplacian segmentation + index window search	Selects frames from contiguous segments and matches OIS/non-OIS using nearby frame indices.
3	Geometric alignment	Feature-based registration	Verifies spatial correspondence between paired images.
4	Motion coherence	Optical flow estimation	Confirms motion consistency and absence of scene changes.
5	Photometric alignment	Intensity mapping	Validates global brightness consistency across the pair.

Table 14. Continued…

#	Validation Stage	Method Used	What It Validates
6	Intensity scaling check	Histogram or mean intensity comparison	Confirms consistent intensity ranges after normalization.
7	Color alignment	Photometric normalization	Reduces global color bias between paired images.
8	Color consistency metric	Mean color comparison in CIELAB space	Verifies chromatic similarity after normalization.
9	Outlier detection	Threshold-based rejection	Removes pairs failing any validation stage.
10	Manual spot check	Visual inspection	Confirms automated validation results.

Training and Testing

After completing data cleaning and multi-stage validation, the finalized datasets will be prepared for model training and evaluation. Since this study benchmarks motion deblurring performance under two hardware stabilization conditions, two separate datasets will be maintained: one captured with OIS and one captured without OIS. Dataset partitioning will be conducted independently for each dataset to ensure that no cross-contamination occurs between stabilization conditions.

For both the OIS and non-OIS datasets, a 60–20–20 distribution will be applied. Sixty percent of each dataset will be allocated to training, twenty percent to validation, and twenty percent to testing. Applying the same splitting strategy to both datasets ensures methodological consistency when comparing stabilization conditions.

The training subsets will be used for domain-adaptive fine-tuning of the selected transformer-based motion deblurring model. The validation subsets will support hyperparameter tuning and monitoring of training behavior. The testing subsets will remain strictly reserved for final performance evaluation during baseline benchmarking and post-fine-tuning benchmarking.

To ensure proportional representation of motion patterns, stratified dataset splitting will be implemented separately for the OIS and non-OIS datasets. Within each stabilization condition, image pairs will first be grouped according to blur generation method: handshake-based, vibration-based, and sliding-based motion. Each group will then be partitioned independently following the 60–20–20 ratio.

As illustrated in Figure 9, for each stabilization condition, 100 image pairs from each blur generation method will be assigned to the training set, resulting in 300 training pairs (60%). The remaining image pairs will be distributed between validation and testing. For the validation set (20%), 34 handshake-based pairs, 33 vibration-based pairs, and 33 sliding-based pairs will be allocated, resulting in 100 validation pairs. The same distribution will be applied to the testing set (20%), producing 100 testing pairs per stabilization condition.

This stratified procedure ensures that both the OIS and non-OIS datasets maintain balanced representation of blur generation methods. By preserving the internal distribution of motion patterns within each stabilization condition, the splitting strategy minimizes sampling bias and supports fair comparison between hardware stabilization settings and model adaptation stages.

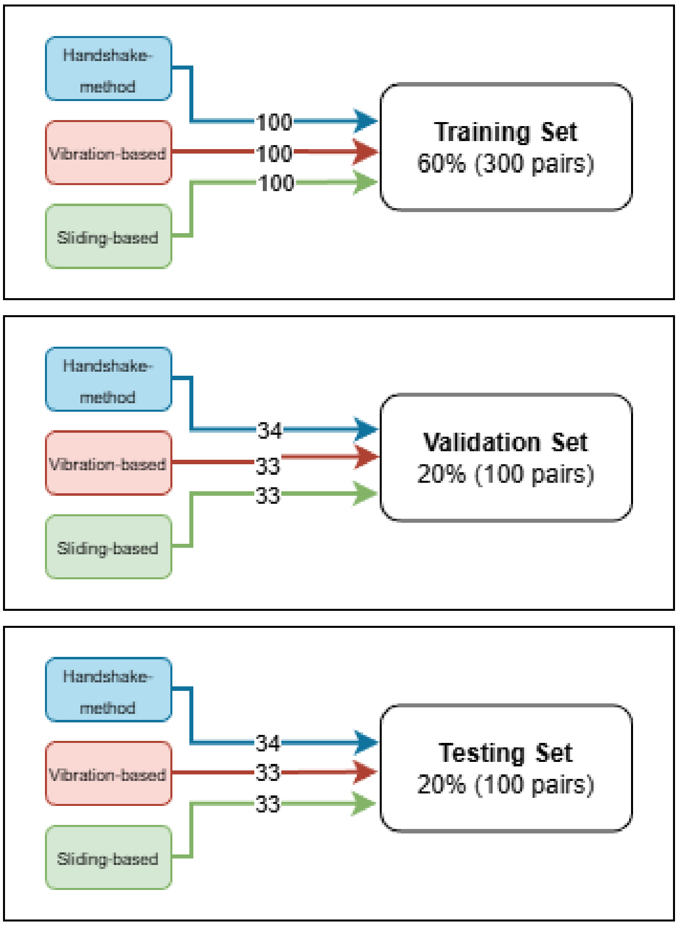

Figure 9. Stratified Dataset Splitting Flow

Data Circulation

Once the dataset has been fully curated, cleaned, validated, and partitioned, it will be prepared for circulation across the subsequent stages of the experimental workflow. The finalized dataset will be organized using a consistent and clearly defined directory structure to support efficient access during model training, baseline benchmarking, post-fine-tuning evaluation, and statistical analysis.

The dataset will be stored locally with redundant backups to ensure data integrity throughout the experimentation phase. In addition, a copy of the finalized dataset will be uploaded to a public Hugging Face repository, which will serve as the primary platform for dataset hosting and documentation. This platform will be used to maintain version control, preserve the dataset configuration used in the experiments, and support reproducibility of the study.

The repository will include structured metadata, clear descriptions of file naming conventions, and documentation of the data acquisition, cleaning, and validation procedures. These materials will ensure that the dataset can be correctly interpreted and reused under the same experimental assumptions. Circulating the dataset through this approach will support transparency and long-term accessibility while ensuring that the dataset remains consistent and well-maintained throughout the research process.

Phase 2: Baseline Benchmarking

Phase 2 will establish the baseline performance of the pre-trained MLWNet model on real-world motion-blurred images captured under optical image stabilization and non-optical image stabilization conditions. This phase will serve as the reference point for evaluating the effect of model fine-tuning in later stages.

Baseline benchmarking will be conducted using the testing subset generated during dataset partitioning. This subset will contain validated paired motion-blurred and sharp reference images from both OIS and non-OIS conditions.

Each motion-blurred image will be processed using the pre-trained MLWNet model without any modification to its architecture or parameters. Inference will be performed under consistent computational settings, and no test-time augmentation or additional post-processing will be applied.

Restoration performance will be evaluated using PSNR and SSIM. These metrics will be computed by comparing the restored outputs with their corresponding sharp reference images, following the definitions summarized in Table 15. Metric computation will be performed separately for OIS and non-OIS image subsets to allow controlled comparison across stabilization conditions.

Table 15. Performance Evaluation Metrics and Formulas

Metric	Formula	Description
MSE	$\mathrm{MSE}(I,S) = \frac{1}{K}\sum_{k=0}^{K-1}(I_k - S_k)^2$	Measures the average squared pixel-wise difference between the restored image I and the sharp reference image S , where K is the total number of pixels.
PSNR	$\mathrm{PSNR}(I,S) = 10\log_{10}\left(\frac{(2^n-1)^2}{\mathrm{MSE}(I,S)}\right)$	Quantifies pixel-level reconstruction accuracy derived from MSE, where n denotes the number of bits per pixel.
SSIM	$\mathrm{SSIM}(I,S) = l(I,S)^{\alpha}\, c(I,S)^{\beta}\, s(I,S)^{\gamma}$	Evaluates perceptual similarity between the restored image I and the reference image S based on luminance, contrast, and structure.

The resulting performance scores will be recorded and preserved as the baseline reference for post-fine-tuning benchmarking and statistical analysis.

Phase 3: Model Fine-Tuning

Phase 3 will involve fine-tuning the pre-trained MLWNet model using the validated real-world motion deblurring dataset. The objective of this phase is to adapt the model to the characteristics of motion blur observed in smartphone images captured under OIS and non-OIS conditions.

Fine-tuning will be performed using the training and validation subsets defined during dataset partitioning. The training subset will be used to update model parameters, while the validation subset will be used to monitor training behavior and guide hyperparameter control. The testing subset will remain excluded from this phase and will be reserved for evaluation.

The model will be initialized using publicly available pre-trained weights, and the network architecture will remain unchanged throughout the fine-tuning process. Training will follow a supervised learning procedure, where motion-blurred images are used as inputs and corresponding sharp reference images are used as targets.

Model optimization will be conducted using fixed and controlled hyperparameter settings. Training will be performed for a limited number of epochs to reduce the risk of overfitting, given the size of the dataset. Validation performance will be monitored to ensure stable convergence.

Upon completion of training, the model state corresponding to the best validation performance will be selected as the final fine-tuned model. The selected model parameters will be fixed and used for post-fine-tuning benchmarking in the next phase.

Hyperparameter Configuration

The hyperparameters used for model fine-tuning are summarized in Table 16. These parameters define the optimization strategy, training settings, and data handling procedures applied throughout the fine-tuning process. All hyperparameters will be fixed prior to training and will remain unchanged to ensure methodological consistency and reproducibility. This controlled configuration ensures that performance differences observed in later phases can be attributed to model adaptation rather than variations in training settings.

Table 16. Hyperparameter Configuration for Model Fine-Tuning

Hyperparameter	Setting	Description
Model	MLWNet (pre-trained)	Pre-trained weights will be used for initialization
Optimizer	AdamW	Adaptive optimizer with decoupled weight decay

Initial learning rate	1×10^{-3}	Controls the magnitude of parameter updates
Learning rate schedule	Cosine annealing	Gradually reduces the learning rate during training
Batch size	Hardware-dependent	Selected based on available GPU memory
Patch size	256 × 256	Fixed input patch size for training samples
Data augmentation	Random flips and rotations	Increases data diversity while preserving blur characteristics
Normalization	Model-defined normalization layers	Stabilizes feature distributions during training
Number of epochs	Fixed and limited	Prevents overfitting on a moderate-sized dataset
Weight initialization	Pre-trained weights	Enables domain-adaptive fine-tuning

Phase 4: Post-Fine-Tuning Benchmarking

Phase 4 will evaluate the performance of the fine-tuned MLWNet model on real-world motion-blurred images captured under OIS and non-OIS conditions. This phase is intended to measure the effect of domain-adaptive fine-tuning by comparing post-fine-tuning performance against the baseline results obtained in Phase 2.

Post-fine-tuning benchmarking will be conducted using the same testing subset used during baseline benchmarking. No additional data will be introduced at this stage. This controlled setup will ensure that all comparisons are performed under identical data and environmental conditions.

Each motion-blurred image in the testing subset will be processed using the fine-tuned model without any further modification or adaptation. Inference will be executed under the same computational configuration used during baseline benchmarking. No test-time augmentation, parameter adjustment, or additional post-processing will be applied. This ensures that any observed performance difference can be attributed solely to the fine-tuning process.

Restoration performance will be evaluated using PSNR and SSIM. These metrics will be computed by comparing the restored image output with its corresponding sharp reference image. MSE will serve as the intermediate computation for PSNR calculation. The definitions and formulas of these metrics are summarized in Table 17.

Metric computation will be performed separately for OIS and non-OIS image subsets. This separation will allow controlled analysis of the interaction between hardware stabilization and model adaptation.

Table 17. Performance Evaluation Metrics and Formulas

Metric	Formula	Description
MSE	$\mathrm{MSE}(I,S)=\frac{1}{K}\sum_{k=0}^{K-1}(I_k-S_k)^2$	Measures the average squared pixel-wise difference between the restored image I and the sharp reference image S , where K is the total number of pixels.
PSNR	$\mathrm{PSNR}(I,S)=10\log_{10}\left(\frac{(2^n-1)^2}{\mathrm{MSE}(I,S)}\right)$	Quantifies pixel-level reconstruction accuracy derived from MSE, where n denotes the number of bits per pixel.
SSIM	$\mathrm{SSIM}(I,S)=l(I,S)^\alpha\,c(I,S)^\beta\,s(I,S)^\gamma$	Evaluates perceptual similarity between the restored image I and the reference image S based on luminance, contrast, and structure.

Performance Evaluation and Comparison Criteria. To explicitly quantify the impact of fine-tuning, performance improvement will be measured using absolute change and percent change.

Absolute change will be computed as:

$$\Delta PSNR = PSNR_{fine-tuned} - PSNR_{pre-trained}$$

$$\Delta SSIM = SSIM_{fine-tuned} - SSIM_{pre-trained}$$

Percent change will be computed as:

$$\%\text{Improvement} = \frac{\text{Fine-tuned} - \text{Pre-trained}}{\text{Pre-trained}} \times 100$$

Mean absolute and percent changes will be computed separately under OIS and non-OIS conditions. The decision criteria that will be used to determine the effectiveness of fine-tuning are summarized in Table 18. These criteria define the statistical, directional, and practical requirements that must be satisfied before concluding that model adaptation is effective.

Table 18. Decision Criteria for Evaluating Fine-Tuning Effectiveness

Criterion	Decision Rule
Statistical Significance	Main effect of model adaptation must be significant at $\alpha = 0.05$.
Direction of Improvement	Mean absolute gain ($\Delta PSNR$ or $\Delta SSIM$) must be positive.
Practical Significance	Partial eta squared ($\eta^2 p$) must indicate at least a small effect ($\eta^2 p \geq 0.01$).
Stabilization Dependence	If interaction is significant, effectiveness will be evaluated separately for OIS and non-OIS conditions.
Metric Consistency	Conclusions will be reported independently for PSNR and SSIM. Improvement in one metric will not imply improvement in the other.

Fine-tuning will be considered effective when statistical significance is achieved, practical effect size is at least small, and mean improvement is positive. If statistical significance is observed but effect size is negligible, the improvement will be interpreted as statistically detectable but practically limited.

If improvement is observed in PSNR but not in SSIM, interpretation will distinguish between pixel-level reconstruction enhancement and perceptual similarity enhancement. Conclusions will be reported separately for each metric.

If a significant interaction effect between stabilization condition and model adaptation is detected, additional simple main effects analysis will be conducted in Phase 5 to determine whether performance improvement is stabilization-dependent.

The statistical procedures used to validate these decision rules are described in the subsequent phase.

Phase 5: Statistical Analysis

Phase 5 will evaluate whether hardware stabilization and domain-adaptive fine-tuning significantly affect motion deblurring performance. Statistical analyses will be conducted separately for PSNR and SSIM to ensure independent interpretation of pixel-level fidelity and perceptual similarity.

Experimental Design and Evaluation Matrix. The experimental design will follow a 2 × 2 factorial structure using Two-Way ANOVA. The first factor will be hardware stabilization conditions, which will consist of two levels: images captured with OIS and images captured without OIS. The second factor will be model adaptation conditions, which will consist of two levels: pre-trained model and fine-tuned model.

This factorial structure will produce four experimental groups. The 2 × 2 evaluation matrix is presented in Table 19.

Table 19. 2 × 2 Evaluation Matrix of Experimental Conditions

Hardware Stabilization	Pre-Trained Model	Fine-Tuned Model
With OIS	Group 1	Group 2

Without OIS	Group 3	Group 4

Group 1 will represent images captured with OIS and restored using the pre-trained model. Group 2 will represent images captured with OIS and restored using the fine-tuned model. Group 3 will represent images captured without OIS and restored using the pre-trained model. Group 4 will represent images captured without OIS and restored using the fine-tuned model.

Each experimental group will generate performance values for PSNR and SSIM, which will serve as input for statistical analysis.

Separate Statistical Analysis for PSNR and SSIM. Two separate Two-Way ANOVA procedures will be conducted. The first analysis will evaluate PSNR across the four experimental groups. The second analysis will evaluate SSIM across the same groups. Conducting separate analyses will ensure that each performance metric is interpreted independently.

Statistical Model. For each performance metric, the two-way ANOVA model will be expressed as:

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

where:

$$\begin{aligned} Y_{ij} &: \text{Observed performance metric (PSNR or SSIM)} \\ \mu &: \text{Overall mean} \\ \alpha_i &: \text{Effect of hardware stabilization condition} \\ \beta_j &: \text{Effect of model adaptation condition} \\ (\alpha\beta)_{ij} &: \text{Interaction effect between the two factors} \\ \varepsilon_{ijk} &: \text{Random error term} \end{aligned}$$

The statistical significance of the main effects and the interaction effect will be evaluated using the following F-ratios:

$$F_A = \frac{MS_A}{MS_E}, \quad F_B = \frac{MS_B}{MS_E}, \quad F_{AB} = \frac{MS_{AB}}{MS_E}$$

where MS_A represents the mean square for hardware stabilization, MS_B represents the mean square for model adaptation, and MS_{AB} represents the mean square for the interaction effect, and MS_E represents the mean square error. Statistical significance will be determined at a significance level of $\alpha = 0.05$.

Hypothesis Testing Structure. The null and alternative hypotheses that will be tested for each metric are summarized in Table 20.

Table 20. Hypothesis Testing per Performance Metric

Effect	Null Hypothesis	Alternative Hypothesis
Hardware Stabilization	There is no statistically significant difference between OIS and non-OIS conditions.	There is a statistically significant difference between OIS and non-OIS conditions.
Model Adaptation	Domain-adaptive fine-tuning does not significantly change performance compared to the pre-trained model.	Domain-adaptive fine-tuning significantly changes performance compared to the pre-trained model.
Interaction	There is no interaction effect between stabilization and adaptation.	There is a statistically significant interaction effect between stabilization and adaptation.

Effect Size Estimation. Effect size will be reported using partial eta squared, computed as:

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{error}}}$$

Effect sizes will be interpreted as follows: (a) 0.01 will indicate a small effect; (b) 0.06 will indicate a medium effect; and (c) 0.14 will indicate a large effect.

Effect size values will be reported separately for stabilization, model adaptation, and interaction effects for both PSNR and SSIM.

Assumption Checking. Prior to conducting the two-way ANOVA, the assumptions of normality of residuals, homogeneity of variance, and absence of extreme outliers will be evaluated. Statistical analysis will be conducted using SPSS or R under the General Linear Model framework. Reported results will include F-values, p-values, effect sizes, and confidence intervals for PSNR and SSIM separately.

Materials and Experimental Environment

This study will require specific hardware, software, cloud storage, and data collection setups to support dataset creation, benchmarking, fine-tuning, and evaluation of motion deblurring performance under different hardware stabilization conditions.

Local Computing Hardware. Local hardware will be used for data acquisition, code preparation, and data management, while computationally intensive tasks will be executed in a cloud-based environment. The minimum specifications of the local computing hardware required for the study are summarized in Table 21.

Table 21. Minimum Specifications of Local Computing Hardware

Component	Minimum specification	Purpose
Processor	Intel Core i5 or equivalent	Supports preprocessing, scripting, and cloud interaction
Memory	At least 8 GB RAM	Enables smooth handling of image and video files

Storage	At least 256 GB local storage	Temporary storage for RAW files and extracted frames
Internet connection	Stable broadband connection	Required for Google Colab access and Google Drive synchronization

Smartphone Devices for Data Acquisition. Two smartphone devices will be used for data acquisition. The Xiaomi Redmi Note 13 will represent the non-OIS condition, while the Xiaomi Poco X6 5G will represent the OIS condition. Both devices are from the same manufacturer to reduce variability in software-based image processing. The detailed camera and storage specifications of both devices, including the applied experimental control measures, are presented in Table 22.

Capture parameters will be standardized across both devices. Only the primary wide camera lens will be used. Resolution, frame rate, and exposure settings will be manually controlled to ensure fair comparison between stabilization conditions.

Table 22. Smartphone Camera and Storage Specifications with Experimental Control Measures

Specification	Xiaomi Redmi Note 13 (Non-OIS)	Xiaomi Poco X6 5G (OIS)	Experimental Control / Justification
Optical Image Stabilization	Not available	Available (OIS)	Independent variable of the study
Electronic Image Stabilization	Not specified	Gyro-EIS	EIS will be disabled to isolate the effect of OIS
Main Camera Resolution	108 MP	64 MP	Output resolution will be standardized before evaluation
Sensor Size	1/1.67"	1/2.0"	Declared as limitation; exposure will be manually standardized
Aperture	f/1.7	f/1.8	Shutter speed and ISO will be manually matched

Pixel Size	0.64μm	0.7μm	Not treated as an experimental variable
Video Resolution	1080p@30fps	4K@30fps, 1080p@30/60fps	Both devices will be restricted to 1080p@30fps
Frame Rate	30 fps	30/60 fps	Both devices will use 30 fps only
RAW Image Capture	Supported	Supported	RAW format will preserve sensor-level data
RAW Video Capture	Supported through a mobile application	Supported through a mobile application	RAW video will enable frame extraction without compression artifacts
Manual Exposure Control	To be enabled	To be enabled	Shutter speed, ISO, focus, and white balance will be locked
Storage & Memory	256GB 8GB RAM	256GB 8GB RAM	Devices with at least 256GB storage will be selected to accommodate RAW image and video data

Camera-related specifications were prioritized because the study focuses on motion blur characteristics and hardware stabilization effects. Device specifications unrelated to image formation and stabilization were not treated as experimental variables, as they do not directly influence motion blur behavior.

Software and Cloud Computing Environment. The primary computing environment for model execution, fine-tuning, and evaluation will be Google Colaboratory (Colab). The minimum cloud computing environment specifications used in the study are summarized in Table 23.

Table 23. Cloud Computing Environment Specifications

Component	Specification
Platform	Google Colaboratory
Runtime type	GPU-enabled runtime

GPU	NVIDIA Tesla T4, P100, or equivalent
Memory	Approximately 12–16 GB RAM
Programming language	Python 3.8 or later
Deep learning framework	PyTorch 1.11 or later
GPU acceleration	CUDA and cuDNN (preconfigured)

Supporting libraries will include Torchvision, NumPy, OpenCV, Pillow, scikit-image, and tqdm. Experiments will be implemented using Jupyter notebooks.

For RAW video acquisition, MotionCam Pro Trial will be used. This mobile camera application allows recording of sensor-level video data without aggressive compression or in-camera post-processing, preserving motion blur characteristics required for accurate benchmarking.

Cloud Storage. Google Drive will be used as the primary cloud storage platform for the study. A storage capacity of up to 2 TB will be allocated to store RAW images, RAW video files, extracted frames, model checkpoints, and experimental results. Google Drive will also be used to synchronize data between local devices and the Google Colab environment.

In addition to Google Drive, the curated dataset constructed in this study will be stored and distributed through the Hugging Face dataset repository. After data collection, preprocessing, and organization, the finalized dataset consisting of paired motion-blurred and sharp images captured under OIS and non-OIS conditions will be uploaded to the Hugging Face dataset hub.

Data and Evaluation Resources. The dataset will consist of paired real-world images derived from motion-blurred and sharp frames extracted from RAW video recordings captured simultaneously using the OIS and non-OIS smartphones.

Quantitative evaluation will be performed using PSNR and SSIM to assess motion deblurring performance before and after model fine-tuning.

Ethical Considerations

This study will follow the ethical standards required for undergraduate research in Computer Science at Cavite State University. The research will be classified as an experimental computing study because it will involve controlled image acquisition, model evaluation, and statistical analysis of performance metrics .

The study will not involve human participants. The dataset will consist only of images of inanimate objects and static scenes captured in public or controlled environments. No human subjects, faces, private individuals, or personally identifiable information will be included in the dataset. As such, informed consent from human participants will not be required. Image collection will avoid private property, restricted locations, and sensitive areas to ensure respect for privacy and security.

One mobile device with OIS will be borrowed for image acquisition. The device owner will provide written and explicit permission prior to its use. The device will be used strictly for capturing experimental images required for this study. No personal files, user accounts, or existing media stored on the device will be accessed, copied, modified, or processed.

If the device owner decides to replace or swap the device during the data collection period, a new written permission will be secured before the use of the replacement device. The same data privacy and handling procedures will be strictly followed.

After image acquisition, all experimental images will be securely transferred to the researchers' password-protected storage system and will be permanently deleted from the borrowed device. The researchers will handle the borrowed device responsibly and will return it in its original condition.

All collected images will be used solely for academic and research purposes. The dataset will not be distributed publicly unless permitted by the adviser and compliant with university guidelines. Access to the dataset will be limited to the researchers and the thesis adviser.

To ensure fairness and scientific integrity, camera settings, capture procedures, and environmental conditions will be applied consistently across both hardware stabilization conditions. The presence or absence of OIS will be treated as the only controlled hardware-related variable. No artificial manipulation will be introduced to favor a specific experimental outcome.

All transformer-based models, software libraries, and evaluation tools will be obtained from publicly available and legally accessible sources. Proper acknowledgment will be given to original authors and developers following academic citation standards. No proprietary datasets, restricted software, or unauthorized resources will be used.

Performance metrics, specifically PSNR and SSIM, will be computed using standardized and reproducible procedures. Statistical analyses will be conducted objectively to determine performance differences. No results will be fabricated, altered, or selectively reported. All findings will be presented transparently, including limitations and possible sources of experimental bias.

Finally, the study will comply with the research policies, manuscript standards, and ethical guidelines of Cavite State University. All procedures will be conducted under the supervision of the assigned thesis adviser and will undergo proper review and approval before implementation.

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APPENDIX A

Frequency Measurement of the Vibration Device

I. Objectives:

This experimental procedure aims to:

1. Measure the vibration frequency generated at selected speed settings of the vibration device using an accelerometer sensor.
2. Establish a quantitative relationship between device speed settings and measured vibration frequency in Hertz (Hz)..
3. Ensure consistency and repeatability of the vibration-based motion generation setup

II. Materials

The following materials were used in the vibration frequency measurement:

- Percussive vibration device
- Adjustable clamp mount
- Flat rigid vibration platform
- Rubber isolation feet
- Smartphone with built-in accelerometer sensor
- Frequency analysis application (FFT-based sensor application)
- Spreadsheet software or piece of paper for data recording

III. Experimental Procedure

The vibration device was rigidly mounted on the flat vibration platform using an adjustable clamp to ensure stable contact. Rubber isolation feet were installed beneath the platform to reduce vibration transfer to surrounding surfaces.

A smartphone with a built-in accelerometer sensor and frequency analysis application was placed flat on the vibration platform. The application displayed the dominant vibration frequency in Hertz (Hz).

The device was activated sequentially from Speed Setting 1 to Speed Setting 32. Each selected setting operated for five seconds to allow vibration stabilization before recording.

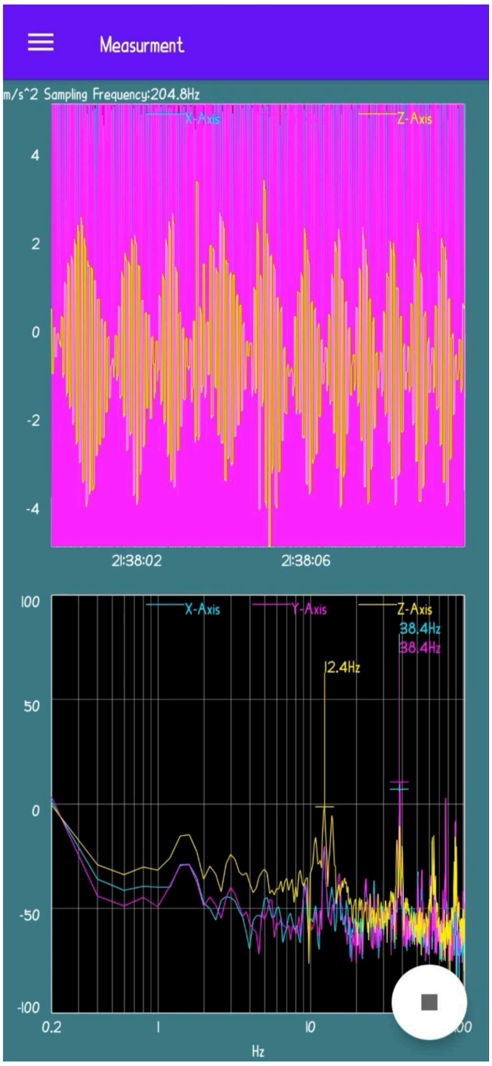
For each setting:

- The dominant vibration frequency in Hertz (Hz) was recorded.
- Environmental conditions were kept stable.
- Platform position and sensor alignment were maintained constant.

A single controlled measurement was conducted per speed setting to maintain consistent experimental configuration.

Vibration-Based Experimental Setup

Sample Frequency Measurement Display in the Sensor Application



IV. Measured Frequency Data

Measured Frequency per Speed Setting

Speed Level	Measured Frequency (Hz)
1	37.7
2	38.4
3	39.4
4	40.6
5	41.7
6	41.7
7	42.9
8	42.9
9	43.9
10	45.1

11	46.3
12	47.4
13	48.5
14	49.6
15	50.6
16	51.5
17	53.1
18	54.1
19	55.1
20	56.3
21	57.3
22	58.5
23	59.4
24	60.5
25	61.5
26	62.3
27	63.5
28	63.5
29	64.5
30	64.5
31	65.1
32	65.2

The results showed a consistent increase in vibration frequency as the device speed setting increased. The minimum recorded frequency was 37.7 Hz at Speed Setting 1, while the maximum recorded frequency was 65.2 Hz at Speed Setting 32.

V. Vibration Frequency Range

Vibration frequency is commonly measured in Hertz (Hz), which represents the number of oscillations per second. The International Organization for Standardization (2016) stated that vibration measurements for most machines generally fall within the range of 10 Hz to 1,000 Hz, depending on the device and operating conditions. Within this broad spectrum, the classification of low-frequency vibration varies across applications. Chen and Liu, as cited in Sun et al. (2022), defined low-frequency vibration as 20 Hz to 200 Hz in brake system analysis, while Thompson, as cited in Sun et al. (2022), identified 2 Hz to 80 Hz as low frequency in railway vibration studies. In energy harvesting research, frequencies from 10 Hz to several hundred Hertz are also regarded as low frequency. In contrast, Hayakawa et al., as cited in Sun et al. (2022), limited ultra-low frequency vibrations to a maximum of 10 Hz when analyzing geomagnetic variations associated with earthquakes. These varying definitions indicate that frequency classifications are context-dependent and not universally fixed, and that ranges above 10 Hz are generally considered within the low-frequency domain in mechanical vibration studies.

VI. Final Frequency-Based Classification

All experimentally measured frequencies fall within the low-frequency vibration range as defined in mechanical vibration literature (2–200 Hz depending on application). Therefore, the device operates entirely within a single vibration category: Low-Frequency Vibration.

Classification	Frequency Range (Hz)
Low-Frequency Vibration	37.7 – 65.2

Conclusion

The accelerometer-based measurement procedure successfully determined the vibration frequency generated by the device across Speed Settings 1 to 32. The recorded frequencies ranged from 37.7 Hz to 65.2 Hz and showed a consistent increasing trend as the speed setting increased. This result established a clear quantitative relationship between device speed levels and measured vibration frequency in Hertz (Hz).

Based on existing vibration literature, the measured frequency range falls entirely within the low-frequency vibration classification used in mechanical vibration studies. The minimum value of 37.7 Hz is above the ultra-low frequency threshold of 10 Hz, while the maximum value of 65.2 Hz remains within the commonly defined low-frequency ranges of 2–80 Hz and 20–200 Hz reported in prior studies. Therefore, the vibration output of the device is classified as Low-Frequency Vibration.

The use of measured frequency values instead of manufacturer-defined speed settings provided an objective and reproducible basis for vibration characterization. The experimentally obtained frequency range of 37.7 Hz to 65.2 Hz establishes a quantitative foundation for subsequent motion blur analysis and ensures methodological consistency in the study.

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APPENDIX B

Device Selection and Decision-Making for
OIS and Non-OIS Devices

Objectives:

This appendix presents the documented process used in selecting the OIS and non-OIS devices for the experiment. Specifically, it aims to:

1. Present the evaluation process for selecting experimental devices.
2. Compare rental, borrowed, and second-hand device options.
3. Justify the final device selection based on cost, availability, technical capability, and experimental repeatability.

Device Selection Criteria

The following criteria were used in selecting the experimental devices:

Criterion	Description
OIS	Presence or absence of hardware-based image stabilization
Video Capability	Minimum 1080p recording support
RAW Capture Support	Support for RAW video output via Camera2 API
Application Compatibility	Compatibility with MotionCam Pro
Hardware Reliability	Stable hardware condition during testing
Budget Feasibility	Cost suitability for repeated experiments
Availability	Continuous access for repeated recording sessions without scheduling restrictions

Rental OIS Phones Considered

Model	Camera Specification	Limitation	Strength	Link
Samsung Galaxy S23 Ultra	**Main Camera:** 200 MP (wide, OIS); 10 MP (telephoto, 3×, OIS); 10 MP (periscope, 10×, OIS); 12 MP (ultrawide). **Video:** 8K@24/30fps; 4K@30/60fps; 1080p@30/60/120/240fps;	• Rental cost for long recordings • Availability depends on rental schedule • ₱2,300 (1 day) • ₱1,600/day (2 days) • ₱1,000/day (4+ days)	Stable flagship OIS baseline reference	https://picmerentph.carrd.co/

	HDR10+; stereo sound; gyro-EIS.			
Samsung Galaxy S24 Ultra	**Main Camera:** 200 MP (wide, OIS); 10 MP (telephoto, 3×, OIS); 50 MP (periscope, 5×, OIS); 12 MP (ultrawide). **Video:** 8K@24/30fps; 4K@30/60/120fps; 1080p@30/60/120/240fps; HDR10+; stereo sound; gyro-EIS.	• Rental cost for long recordings • Availability depends on rental schedule • ₱2,300 (1 day) • ₱1,600/day (2 days) • ₱1,000/day (4+ days)	Improved ISP processing comparison	https://picmerentph.carrd.co/
Samsung Galaxy S25 Ultra	**Main Camera:** 200 MP (wide, OIS); 10 MP (telephoto, 3×, OIS); 50 MP (periscope, 5×, OIS); 50 MP (ultrawide). **Video:** 8K@24/30fps; 4K@30/60/120fps; 1080p@30/60/120/240fps; 10-bit HDR; HDR10+; stereo sound; gyro-EIS.	• Rental cost for long recordings • Availability depends on rental schedule • ₱2,300 (1 day) • ₱1,600/day (2 days) • ₱1,000/day (4+ days)	Best stabilization and newest imaging pipeline	https://picmerentph.carrd.co/

Borrowed OIS Phones Considered

Model	Camera Specification	Limitation	Strength
Xiaomi Redmi Note 13 Pro	**Main Camera:** 200 MP (wide, OIS); 8 MP (ultrawide); 2 MP (macro). **Video:** 4K@30fps; 1080p@30/60/120fps; gyro-EIS.	Available only before March	No rental cost
Xiaomi Poco X6 5G	**Main Camera:** 64 MP (wide, OIS); 8 MP (ultrawide); 2 MP (macro). **Video:** 4K@30fps; 1080p@30/60fps; gyro-EIS.	The owner wants to swap phones	Can be used anytime, allowing repeated recordings without cost

Second-Hand OIS Phones Considered

Model	Camera Specification	Limitation	Strength	Link
Xiaomi Redmi Note 13 Pro	**Main Camera:** 200 MP (wide, OIS); 8 MP (ultrawide); 2 MP (macro). **Video:** 4K@30fps; 1080p@30/60/120fps; gyro-EIS.	• Price: ₱7,800 • Used unit condition may vary	Affordable high-resolution OIS device for extended testing	https://www.facebook.com/marketplace/item/2339482503236764/
Samsung Galaxy S9	**Main Camera:** 12 MP (wide, OIS). **Video:** 4K@30/60fps; 1080p@30/60/240fps; 720p@960fps; HDR; stereo sound; gyro-EIS.	• Price: ₱2,000 • Old hardware; small battery capacity	Provides older stabilization data for model generalization	https://www.facebook.com/marketplace/item/1612571729985225/
Samsung Galaxy Note 8	**Main Camera:** 12 MP (wide, OIS); 12 MP (telephoto, 2×, OIS). **Video:** 4K@30fps; 1080p@30/60fps; 720p@240fps; HDR; stereo sound; gyro-EIS.	• Price: ₱2,000 • Physical defects; aging components	Represents extreme real-world degraded camera conditions	https://www.facebook.com/marketplace/item/1868183540507968/

Non-OIS Phone Considered

Model	Camera Specification	Limitation	Strength
Xiaomi Redmi Note 13	**Main Camera:** 108 MP, f/1.7 (wide), 1/1.67", 0.64µm, PDAF **Ultrawide:** 8 MP, f/2.2 **Auxiliary Lens:** Included **Features:** LED flash, HDR, panorama **Video:** 1080p@30fps	None identified	Available anytime; researcher's own device

Final Device Selection and Justification

Selected OIS Device

The Xiaomi Poco X6 5G was selected as the Optical Image Stabilization (OIS) device for the experiment.

The selection was based on the following reasons:

1. **Presence of Optical Image Stabilization (OIS).** The device includes hardware-based OIS, which is required for comparison against a non-OIS baseline.
2. **RAW Capture Support.** The device supports RAW video output through the Camera2 API, which is necessary for MotionCam Pro.
3. **Application Compatibility.** The device is compatible with the MotionCam Pro application and allows manual camera control during recording.
4. **Video Capability.** It supports 4K and 1080p recording, exceeding the minimum 1080p requirement.
5. **Budget Feasibility.** The device is borrowed and does not require rental cost.
6. **Availability.** It can be used anytime through phone swapping, allowing repeated recordings without scheduling restrictions.
7. **Hardware Condition.** The device is in stable condition and suitable for controlled experimental trials.

Compared to rental flagship devices, the Xiaomi Poco X6 5G provides sufficient technical capability without the financial and scheduling limitations associated with rental options. Compared to second-hand units, it offers better hardware reliability and lower risk of failure during repeated trials.

Selected Non-OIS Device

The Xiaomi Redmi Note 13 was selected as the non-OIS baseline device.

The selection was based on the following reasons:

1. **Absence of Optical Image Stabilization.** The device does not include OIS, making it suitable as the experimental baseline.
2. **Video Capability.** It supports 1080p recording, meeting the minimum requirement.
3. **RAW Capture Support.** The device supports Camera2 API, allowing RAW recording through MotionCam Pro.
4. **Application Compatibility.** It is compatible with the MotionCam Pro application.

5. **Availability.** The device is the researcher’s own phone and is continuously available for repeated testing.

6. **Budget Feasibility.** No additional cost is required for its use.

7. **Hardware Condition.** The device is in stable condition and suitable for controlled experimental trials.

The Xiaomi Redmi Note 13 provides a consistent and cost-efficient non-OIS baseline while ensuring experimental repeatability and uninterrupted access.

Summary of Final Selection

The Xiaomi Poco X6 5G was selected as the OIS device, while the Xiaomi Redmi Note 13 was selected as the non-OIS device. Both devices satisfy the established selection criteria, including RAW support, application compatibility, budget feasibility, and availability for repeated experimental trials. The final selection ensures controlled comparison under identical testing conditions.

APPENDIX C

Device Hardware Specifications Comparison
for OIS and Non-OIS Device

I. Objectives:

This appendix aims to:

1. Present the complete hardware specifications of the OIS and non-OIS devices used in the experiment.

II. Devices Used in the Experiment

- OIS Device: Xiaomi Poco X6 5G
- Non-OIS Device: Xiaomi Redmi Note 13

All specifications are based on official manufacturer documentation

Processor and Performance Specifications

Specification	Xiaomi Poco X6 5G	Xiaomi Redmi Note 13
Processor	Snapdragon® 7s Gen 2 (4nm)	Snapdragon® 685 (6nm)
CPU	Octa-core, up to 2.4GHz	Octa-core, up to 2.8GHz
GPU	Adreno™ GPU	Adreno 610
RAM	8GB / 12GB (LPDDR4X)	6GB / 8GB (LPDDR4X)
Storage	256GB / 512GB (UFS 2.2)	128GB / 256GB (UFS 2.2)
Expandable Storage	Not supported	Up to 1TB (microSD)
Operating System	MIUI 14 for POCO	MIUI 14

Physical and Build Specifications

Specification	Xiaomi Poco X6 5G	Xiaomi Redmi Note 13
Height	161.15 mm	162.24 mm
Width	74.24 mm	75.55 mm
Thickness	7.98 mm	7.97 mm
Weight	181 g	188.5 g
Glass Protection	Gorilla Glass Victus	Gorilla Glass 3
Splash Resistance	IP54	IP54

Display Specifications

Specification	Xiaomi Poco X6 5G	Xiaomi Redmi Note 13

Screen Size	6.67-inch AMOLED	6.67-inch AMOLED
Resolution	2712 × 1220 (1.5K)	2400 × 1080 (FHD+)
Refresh Rate	Up to 120Hz	Up to 120Hz
Peak Brightness	1800 nits	1800 nits
Colo Gamut	100% DCI-P3	100% DCI-P3

Camera System and Video Capabilities

Specification	**Xiaomi Poco X6 5G (OIS)**	**Xiaomi Redmi Note 13 (Non-OIS)**
Main Camera	64MP, f/1.79	108MP, f/1.75
Ultra-Wide	8MP	8MP
Macro	2MP	2MP
Optical Image Stabilization	Yes	No
Maximum Video Resolution	4K at 30fps	1080p at 30fps
1080p Recording	30fps / 60fps	30fps

The Xiaomi Poco X6 5G includes OIS, which mechanically compensates for camera motion. The Xiaomi Redmi Note 13 does not include OIS and serves as the non-stabilized baseline device.

Battery and Charging Specifications

Specification	**Xiaomi Poco X6 5G**	**Xiaomi Redmi Note 13**
Battery Capacity	5100 mAh	5000 mAh
Charging	67W Turbo Charging	33W Fast Charging
Charging Port	USB Type-C	USB Type-C

Higher charging capacity supports extended experimental recording sessions.

Sensors

Sensor	**Xiaomi Poco X6 5G**	**Xiaomi Redmi Note 13**
Accelerometer	Yes	Yes

Gyroscope	Yes	Yes
Ambient Light Sensor	Yes	Yes
Proximity Sensor	Yes	Yes
Electronic Compass	Yes	Yes
IR Blaster	Yes	Yes
X-axis Linear Vibration Motor	Yes	No

Conclusion

The hardware comparison confirms that the primary experimental difference between the two devices is the presence of OIS in the Xiaomi Poco X6 5G and its absence in the Xiaomi Redmi Note 13.

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APPENDIX D

Individual Reflection on Learning

INDIVIDUAL REFLECTION ON LEARNING

The development of our thesis outline helped me realize how important planning, organization, and teamwork are in conducting research. During the process, I learned that properly scheduling tasks and making sure that all members are available during meetings whether online or face-to-face is important to keep the work progressing smoothly. I also realized that the contents of Chapters 1 to 3 should always be aligned with each other, especially the Review of Related Literature and Review of Related Studies, since these sections support the objectives and methodology of the research. Another important lesson I learned is the value of time management in thesis writing. Even small daily progress can help reduce pressure and prevent burnout among group members.

My main contribution during the development of the thesis outline was assisting in documentation, particularly supporting the documentation head in preparing Chapters 1 to 3. I helped by searching for reliable literature and related studies relevant to our research topic. I also served as the 3D modeler responsible for creating the models of the experimental setups that will be used in the methodology during the development phase. In addition, I acted as the main point of communication between our thesis adviser and our technical critic. I was responsible for messaging them online, sending documents through email, and updating them about our progress. I also helped with proofreading the outline to ensure that the contents were consistent and aligned across the chapters. During meetings and consultations, I took notes and prepared the Minutes of the Meeting (MoM) to document discussions and important decisions.

In terms of the quality of my work, I made sure that the tasks assigned to me were completed on or before the deadline. I also allowed my groupmates to review my work before final submission because I believe that feedback helps improve the quality of our thesis. Since I am not always confident that my work is perfect, I value the suggestions and corrections from my teammates to ensure that our final output is accurate and well-prepared.

For communication, our group used several methods to coordinate tasks and discussions. We used Messenger for quick updates, Discord for online meetings, and frequently held face-to-face meetings after class or sometimes at one of our houses so we could work on the thesis together. These communication methods helped us stay connected and continue making progress even when our schedules were different.

In terms of collaboration with my team, including our adviser and technical critic, our group frequently held face-to-face meetings compared to other groups. My main role during meetings was taking notes to ensure that important discussions, suggestions, and instructions were properly recorded. I also handled communication with our adviser and technical critic by sending documents and updates through email and attending scheduled meetings to record their feedback.

One of the main challenges during the development of our thesis outline was aligning our schedules since my groupmates had similar class schedules while mine was different. Because of this, we needed to adjust our available time so we could still meet and work together. Another challenge for me was communication, since I sometimes find it difficult to express my ideas clearly during discussions. Balancing thesis work with my other academic responsibilities was also challenging. To overcome these difficulties, I tried to improve my time management and adjust my schedule whenever needed.

Overall, my thesis journey has been a valuable learning experience. It helped me develop skills in teamwork, communication, time management, and research documentation. Despite the challenges we faced, working together allowed our group to support each other and continue progressing in our thesis. This experience helped me grow not only as a student researcher but also as a responsible team member.

JUSTIN CHRISTIAN A. DIOKNO

INDIVIDUAL REFLECTION ON LEARNING

Learning about thesis outline development helped me understand that research in Computer Science should begin with a clearly defined research problem. Before this, I used to think that starting with a title or a solution was acceptable. However, I learned that the proper process begins with identifying a research gap, then formulating objectives, and only after that defining the title and proposed computing solution. This structured approach made our outline more logical and aligned.

I also realized that developing the thesis outline is not just about writing Chapters 1 to 3. It requires careful alignment between the research problem, objectives, methodology, and expected results. Each section must support one another. If the objectives are not clear, the methodology will also be unclear. This experience made me more careful in planning each part of the outline and ensuring consistency throughout the document.

Another important lesson I learned is the importance of reviewing related literature properly. It is not enough to summarize previous studies. It is necessary to identify limitations, research gaps, and unresolved issues. This process strengthens the justification of the proposed study and proves its relevance and necessity.

I also learned that feasibility and scope are very important in undergraduate research. The study must be realistic in terms of time, tools, available data, and team capability. A good outline does not only present a good idea, but also demonstrates that the idea can be implemented within the given timeframe and constraints.

As the team leader and documentation head, I realized that thesis outline development is both an academic and leadership responsibility. My role was not only to compile and organize the document, but also to guide the team, monitor progress, and maintain alignment between sections. During this process, I created a simple formula to guide our team: Success = Communication + Time Management.

I emphasized open and continuous communication through face-to-face discussions and regular updates. I had already raised concerns about improving communication because I noticed that it directly affects productivity and collaboration. Clear communication improves confidence, teamwork, and motivation, and it helps resolve misunderstandings more quickly.

For time management, I encouraged structured meeting schedules to avoid procrastination and to simulate a professional working environment. Discipline in scheduling helped us stay organized and focused on our tasks.

I also realized that no group is perfect. Conflict will still arise no matter how organized or prepared the team is. Differences in opinions, working styles, and personal constraints are natural. However, what truly matters is the capability to handle such situations properly. I recognized that our ability to manage conflicts and maintain professionalism still needs improvement. Leadership is not about avoiding conflict, but about addressing it constructively and guiding the team toward solutions.

Overall, learning about thesis outline development improved my understanding of structured research writing and teamwork. It trained me to think critically, plan systematically, justify decisions logically, and lead responsibly. More importantly, it made me realize that research is a continuous process that requires discipline, collaboration, maturity, and constant self-improvement.

JOSSEL JOHN S. DUMAOP

INDIVIDUAL REFLECTION ON LEARNING

Working on our thesis, “Benchmarking Motion Deblurring under Optical Image Stabilization Using a Transformer-Based Model,” has been both challenging and rewarding for me as a BSCS student. In our thesis outline development, I served as the group’s machine learning engineer.

My primary responsibility in the team was to research and consolidate technical information for our paper, particularly in the methodology section. I ensured that complex concepts related to motion deblurring, Optical Image Stabilization (OIS), and transformer-based models were clearly explained to my teammates so that everyone could understand the process of our study. I also served as the group’s proofreader, carefully reviewing our drafts to maintain clarity, coherence, and academic quality. One of my major contributions was designing the system’s process flow, which served as an essential guide for implementing our study. However, I recognize that I still need to improve in creating more structured and detailed guides to further help my team clearly visualize each step of the research process.

In terms of work quality and independence, I consistently completed my assigned tasks without requiring supervision. While our team occasionally needed coordination and clarification on certain tasks, I took initiative to ensure that my responsibilities were fulfilled on time. Our group was also recognized for frequently meeting face-to-face compared to other groups, and we supplemented this with regular online meetings. I was often the one communicating with our technical adviser and technical critic, either by directly messaging them or by validating the messages prepared by my teammates. I also took the lead in interacting with them and other professors during face-to-face consultations. These experiences strengthened my professional communication skills and confidence in academic discussions.

Our team maintained strong timeliness and reliability throughout the thesis outline phase. We consistently met deadlines for revisions and assigned tasks, and we implemented daily progress reports to monitor our work. However, one of the major challenges I encountered was the frequency and duration of meetings, which sometimes overlapped with my personal responsibilities outside academics. In many instances, time management alone was not sufficient because of the fixed and overlapping schedules. Despite this, I remained committed to fulfilling my role by adjusting my routine and prioritizing critical tasks.

Overall, this thesis journey significantly enhanced my ability to locate and evaluate crucial information from academic literature. Compared to before, I am now more confident in identifying the specific technical details needed to support our study and justify our methodology. The knowledge I gained about motion deblurring, OIS, and transformer-based models became one of the key factors that enabled our team to construct a solid research paper. More importantly, this experience strengthened my independence and sense of responsibility as a researcher. Moving forward, I aim to further improve my ability to create clearer research guides for my team and continue developing both my technical and collaborative skills.

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